MATH 547 Week 2 Reading Packet

Source excerpts

Source excerpt	Packet pp.
Shalev-Shwartz and Ben-David, *Understanding Machine Learning*	
Chapter 5, Sections 5.1–5.2 (Theorem 5.1); printed pp. 37–41.	
No-Free-Lunch, the failure of universal learning, and the approximation–estimation decomposition.	2–6
Shalev-Shwartz and Ben-David, *Understanding Machine Learning*	
Chapter 6, Sections 6.1–6.5.1; printed pp. 43–50.	
Threshold classes, shattering, VC dimension, geometric examples, growth functions, and Sauer's lemma.	7–14
Giné and Nickl, *Mathematical Foundations of Infinite-Dimensional Statistical Models*	
Chapter 3, Section 3.6 (Theorem 3.6.2); printed pp. 212–227.	
VC set classes, Sauer's bound, permanence properties, and examples.	15–30

All printed page numbers refer to the pagination in the original sources. Packet page numbers count this guide as page 1.

is that there exists h in some predefined hypothesis class $\mathcal{H}$, such that $L_{\mathcal{D}}(h) = 0$. A softer type of prior knowledge on $\mathcal{D}$ is assuming that $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$ is small. In a sense, this weaker assumption on $\mathcal{D}$ is a prerequisite for using the agnostic PAC model, in which we require that the risk of the output hypothesis will not be much larger than $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$.

In the second part of this chapter we study the benefits and pitfalls of using a hypothesis class as a means of formalizing prior knowledge. We decompose the error of an ERM algorithm over a class $\mathcal{H}$ into two components. The first component reflects the quality of our prior knowledge, measured by the minimal risk of a hypothesis in our hypothesis class, $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$. This component is also called the *approximation error*, or the *bias* of the algorithm toward choosing a hypothesis from $\mathcal{H}$. The second component is the error due to overfitting, which depends on the size or the complexity of the class $\mathcal{H}$ and is called the *estimation error*. These two terms imply a tradeoff between choosing a more complex $\mathcal{H}$ (which can decrease the bias but increases the risk of overfitting) or a less complex $\mathcal{H}$ (which might increase the bias but decreases the potential overfitting).

5.1 THE NO-FREE-LUNCH THEOREM

In this part we prove that there is no universal learner. We do this by showing that no learner can succeed on all learning tasks, as formalized in the following theorem:

Theorem 5.1. (No-Free-Lunch) *Let A be any learning algorithm for the task of binary classification with respect to the 0–1 loss over a domain $\mathcal{X}$. Let m be any number smaller than $|\mathcal{X}|/2$, representing a training set size. Then, there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$ such that:*

1. *There exists a function $f : \mathcal{X} \to \{0,1\}$ with $L_{\mathcal{D}}(f) = 0$.*
2. *With probability of at least $1/7$ over the choice of $S \sim \mathcal{D}^m$ we have that $L_{\mathcal{D}}(A(S)) \geq 1/8$.*

This theorem states that for every learner, there exists a task on which it fails, even though that task can be successfully learned by another learner. Indeed, a trivial successful learner in this case would be an ERM learner with the hypothesis class $\mathcal{H} = \{f\}$, or more generally, ERM with respect to any finite hypothesis class that contains f and whose size satisfies the equation $m \geq 8 \log(7|\mathcal{H}|/6)$ (see Corollary 2.3).

Proof. Let C be a subset of $\mathcal{X}$ of size $2m$. The intuition of the proof is that any learning algorithm that observes only half of the instances in C has no information on what should be the labels of the rest of the instances in C. Therefore, there exists a "reality," that is, some target function f, that would contradict the labels that $A(S)$ predicts on the unobserved instances in C.

Note that there are $T = 2^{2m}$ possible functions from C to $\{0,1\}$. Denote these functions by $f_1, \ldots, f_T$. For each such function, let $\mathcal{D}_i$ be a distribution over $C \times \{0,1\}$ defined by

$$\mathcal{D}_i(\{(x,y)\}) = \begin{cases} 1/|C| & \text{if } y = f_i(x) \\ 0 & \text{otherwise.} \end{cases}$$

That is, the probability to choose a pair (x, y) is $1/|C|$ if the label y is indeed the true label according to f_i, and the probability is 0 if $y \neq f_i(x)$. Clearly, $L_{\mathcal{D}_i}(f_i) = 0$.

We will show that for every algorithm, A, that receives a training set of m examples from $C \times \{0,1\}$ and returns a function $A(S) : C \to \{0,1\}$, it holds that

$$\max_{i \in [T]} \mathop{\mathbb{E}}_{S \sim \mathcal{D}_i^m} [L_{\mathcal{D}_i}(A(S))] \geq 1/4. \tag{5.1}$$

Clearly, this means that for every algorithm, A', that receives a training set of m examples from $\mathcal{X} \times \{0,1\}$ there exist a function $f : \mathcal{X} \to \{0,1\}$ and a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$, such that $L_{\mathcal{D}}(f) = 0$ and

$$\mathop{\mathbb{E}}_{S \sim \mathcal{D}^m} [L_{\mathcal{D}}(A'(S))] \geq 1/4. \tag{5.2}$$

It is easy to verify that the preceding suffices for showing that $\mathbb{P}[L_{\mathcal{D}}(A'(S)) \geq 1/8] \geq 1/7$, which is what we need to prove (see Exercise 5.1).

We now turn to proving that Equation (5.1) holds. There are $k = (2m)^m$ possible sequences of m examples from C. Denote these sequences by $S_1, \ldots, S_k$. Also, if $S_j = (x_1, \ldots, x_m)$ we denote by S_j^i the sequence containing the instances in S_j labeled by the function f_i, namely, $S_j^i = ((x_1, f_i(x_1)), \ldots, (x_m, f_i(x_m)))$. If the distribution is $\mathcal{D}_i$ then the possible training sets A can receive are $S_1^i, \ldots, S_k^i$, and all these training sets have the same probability of being sampled. Therefore,

$$\mathop{\mathbb{E}}_{S \sim \mathcal{D}_i^m} [L_{\mathcal{D}_i}(A(S))] \;=\; \frac{1}{k} \sum_{j=1}^{k} L_{\mathcal{D}_i}(A(S_j^i)). \tag{5.3}$$

Using the facts that "maximum" is larger than "average" and that "average" is larger than "minimum," we have

$$\begin{aligned} \max_{i \in [T]} \frac{1}{k} \sum_{j=1}^{k} L_{\mathcal{D}_i}(A(S_j^i)) &\geq \frac{1}{T} \sum_{i=1}^{T} \frac{1}{k} \sum_{j=1}^{k} L_{\mathcal{D}_i}(A(S_j^i)) \\ &= \frac{1}{k} \sum_{j=1}^{k} \frac{1}{T} \sum_{i=1}^{T} L_{\mathcal{D}_i}(A(S_j^i)) \\ &\geq \min_{j \in [k]} \frac{1}{T} \sum_{i=1}^{T} L_{\mathcal{D}_i}(A(S_j^i)). \end{aligned} \tag{5.4}$$

Next, fix some $j \in [k]$. Denote $S_j = (x_1, \ldots, x_m)$ and let $v_1, \ldots, v_p$ be the examples in C that do not appear in S_j. Clearly, $p \geq m$. Therefore, for every function $h : C \to$

$\{0,1\}$ and every i we have

$$\begin{aligned} L_{\mathcal{D}_i}(h) &= \frac{1}{2m}\sum_{x\in C}\mathbb{1}_{[h(x)\neq f_i(x)]} \\ &\geq \frac{1}{2m}\sum_{r=1}^{p}\mathbb{1}_{[h(v_r)\neq f_i(v_r)]} \\ &\geq \frac{1}{2p}\sum_{r=1}^{p}\mathbb{1}_{[h(v_r)\neq f_i(v_r)]}. \end{aligned} \tag{5.5}$$

Hence,

$$\begin{aligned} \frac{1}{T}\sum_{i=1}^{T} L_{\mathcal{D}_i}(A(S_j^i)) &\geq \frac{1}{T}\sum_{i=1}^{T}\frac{1}{2p}\sum_{r=1}^{p}\mathbb{1}_{[A(S_j^i)(v_r)\neq f_i(v_r)]} \\ &= \frac{1}{2p}\sum_{r=1}^{p}\frac{1}{T}\sum_{i=1}^{T}\mathbb{1}_{[A(S_j^i)(v_r)\neq f_i(v_r)]} \\ &\geq \frac{1}{2}\cdot\min_{r\in[p]}\frac{1}{T}\sum_{i=1}^{T}\mathbb{1}_{[A(S_j^i)(v_r)\neq f_i(v_r)]}. \end{aligned} \tag{5.6}$$

Next, fix some $r\in[p]$. We can partition all the functions in $f_1,\ldots,f_T$ into $T/2$ disjoint pairs, where for a pair $(f_i, f_{i'})$ we have that for every $c\in C$, $f_i(c)\neq f_{i'}(c)$ if and only if $c=v_r$. Since for such a pair we must have $S_j^i = S_j^{i'}$, it follows that

$$\mathbb{1}_{[A(S_j^i)(v_r)\neq f_i(v_r)]} + \mathbb{1}_{[A(S_j^{i'})(v_r)\neq f_{i'}(v_r)]} = 1,$$

which yields

$$\frac{1}{T}\sum_{i=1}^{T}\mathbb{1}_{[A(S_j^i)(v_r)\neq f_i(v_r)]} = \frac{1}{2}.$$

Combining this with Equation (5.6), Equation (5.4), and Equation (5.3), we obtain that Equation (5.1) holds, which concludes our proof. □

5.1.1 No-Free-Lunch and Prior Knowledge

How does the No-Free-Lunch result relate to the need for prior knowledge? Let us consider an ERM predictor over the hypothesis class $\mathcal{H}$ of all the functions f from X to $\{0,1\}$. This class represents lack of prior knowledge: Every possible function from the domain to the label set is considered a good candidate. According to the No-Free-Lunch theorem, any algorithm that chooses its output from hypotheses in $\mathcal{H}$, and in particular the ERM predictor, will fail on some learning task. Therefore, this class is not PAC learnable, as formalized in the following corollary:

Corollary 5.2. *Let $\mathcal{X}$ be an infinite domain set and let $\mathcal{H}$ be the set of all functions from $\mathcal{X}$ to $\{0,1\}$. Then, $\mathcal{H}$ is not PAC learnable.*

Proof. Assume, by way of contradiction, that the class is learnable. Choose some $\epsilon < 1/8$ and $\delta < 1/7$. By the definition of PAC learnability, there must be some

learning algorithm A and an integer $m = m(\epsilon, \delta)$, such that for any data-generating distribution over $\mathcal{X} \times \{0,1\}$, if for some function $f : \mathcal{X} \to \{0,1\}$, $L_{\mathcal{D}}(f) = 0$, then with probability greater than $1 - \delta$ when A is applied to samples S of size m, generated i.i.d. by $\mathcal{D}$, $L_{\mathcal{D}}(A(S)) \leq \epsilon$. However, applying the No-Free-Lunch theorem, since $|\mathcal{X}| > 2m$, for every learning algorithm (and in particular for the algorithm A), there exists a distribution $\mathcal{D}$ such that with probability greater than $1/7 > \delta$, $L_{\mathcal{D}}(A(S)) > 1/8 > \epsilon$, which leads to the desired contradiction. □

How can we prevent such failures? We can escape the hazards foreseen by the No-Free-Lunch theorem by using our prior knowledge about a specific learning task, to avoid the distributions that will cause us to fail when learning that task. Such prior knowledge can be expressed by restricting our hypothesis class.

But how should we choose a good hypothesis class? On the one hand, we want to believe that this class includes the hypothesis that has no error at all (in the PAC setting), or at least that the smallest error achievable by a hypothesis from this class is indeed rather small (in the agnostic setting). On the other hand, we have just seen that we cannot simply choose the richest class – the class of all functions over the given domain. This tradeoff is discussed in the following section.

5.2 ERROR DECOMPOSITION

To answer this question we decompose the error of an $\text{ERM}_{\mathcal{H}}$ predictor into two components as follows. Let h_S be an $\text{ERM}_{\mathcal{H}}$ hypothesis. Then, we can write

$$L_{\mathcal{D}}(h_S) = \epsilon_{\text{app}} + \epsilon_{\text{est}} \quad \text{where}: \ \epsilon_{\text{app}} = \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h), \quad \epsilon_{\text{est}} = L_{\mathcal{D}}(h_S) - \epsilon_{\text{app}}. \tag{5.7}$$

- **The Approximation Error** – the minimum risk achievable by a predictor in the hypothesis class. This term measures how much risk we have because we restrict ourselves to a specific class, namely, how much *inductive bias* we have. The approximation error does not depend on the sample size and is determined by the hypothesis class chosen. Enlarging the hypothesis class can decrease the approximation error.

  Under the realizability assumption, the approximation error is zero. In the agnostic case, however, the approximation error can be large.[1]
- **The Estimation Error** – the difference between the approximation error and the error achieved by the ERM predictor. The estimation error results because the empirical risk (i.e., training error) is only an estimate of the true risk, and so the predictor minimizing the empirical risk is only an estimate of the predictor minimizing the true risk.

  The quality of this estimation depends on the training set size and on the size, or complexity, of the hypothesis class. As we have shown, for a finite hypothesis class, ϵ_{est} increases (logarithmically) with $|\mathcal{H}|$ and decreases with m. We can

[1] In fact, it always includes the error of the Bayes optimal predictor (see Chapter 3), the minimal yet inevitable error, because of the possible nondeterminism of the world in this model. Sometimes in the literature the term *approximation error* refers not to $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$, but rather to the excess error over that of the Bayes optimal predictor, namely, $\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) - \epsilon_{\text{Bayes}}$.

think of the size of $\mathcal{H}$ as a measure of its complexity. In future chapters we will define other complexity measures of hypothesis classes.

Since our goal is to minimize the total risk, we face a tradeoff, called the *bias-complexity tradeoff*. On one hand, choosing $\mathcal{H}$ to be a very rich class decreases the approximation error but at the same time might increase the estimation error, as a rich $\mathcal{H}$ might lead to *overfitting*. On the other hand, choosing $\mathcal{H}$ to be a very small set reduces the estimation error but might increase the approximation error or, in other words, might lead to *underfitting*. Of course, a great choice for $\mathcal{H}$ is the class that contains only one classifier – the Bayes optimal classifier. But the Bayes optimal classifier depends on the underlying distribution $\mathcal{D}$, which we do not know (indeed, learning would have been unnecessary had we known $\mathcal{D}$).

Learning theory studies how rich we can make $\mathcal{H}$ while still maintaining reasonable estimation error. In many cases, empirical research focuses on designing good hypothesis classes for a certain domain. Here, "good" means classes for which the approximation error would not be excessively high. The idea is that although we are not experts and do not know how to construct the optimal classifier, we still have some prior knowledge of the specific problem at hand, which enables us to design hypothesis classes for which both the approximation error and the estimation error are not too large. Getting back to our papayas example, we do not know how exactly the color and hardness of a papaya predict its taste, but we do know that papaya is

rectangle in the color-hardness space may be a good predictor.

5.3 SUMMARY

The No-Free-Lunch theorem states that there is no universal learner. Every learner has to be specified to some task, and use some prior knowledge about that task, in order to succeed. So far we have modeled our prior knowledge by restricting our output hypothesis to be a member of a chosen hypothesis class. When choosing this hypothesis class, we face a tradeoff, between a larger, or more complex, class that is more likely to have a small approximation error, and a more restricted class that would guarantee that the estimation error will be small. In the next chapter we will study in more detail the behavior of the estimation error. In Chapter 7 we will discuss alternative ways to express prior knowledge.

5.4 BIBLIOGRAPHIC REMARKS

(Wolpert & Macready 1997) proved several no-free-lunch theorems for optimization, but these are rather different from the theorem we prove here. The theorem we prove here is closely related to lower bounds in VC theory, as we will study in the next chapter.

5.5 EXERCISES

5.1 Prove that Equation (5.2) suffices for showing that $\mathbb{P}[L_{\mathcal{D}}(A(S)) \geq 1/8] \geq 1/7$.
Hint: Let θ be a random variable that receives values in $[0, 1]$ and whose expectation satisfies $\mathbb{E}[\theta] \geq 1/4$. Use Lemma B.1 to show that $\mathbb{P}[\theta \geq 1/8] \geq 1/7$.

6

The VC-Dimension

In the previous chapter, we decomposed the error of the $\text{ERM}_{\mathcal{H}}$ rule into approximation error and estimation error. The approximation error depends on the fit of our prior knowledge (as reflected by the choice of the hypothesis class $\mathcal{H}$) to the underlying unknown distribution. In contrast, the definition of PAC learnability requires that the estimation error would be bounded uniformly over all distributions.

Our current goal is to figure out which classes $\mathcal{H}$ are PAC learnable, and to characterize exactly the sample complexity of learning a given hypothesis class. So far we have seen that finite classes are learnable, but that the class of all functions (over an infinite size domain) is not. What makes one class learnable and the other unlearnable? Can infinite-size classes be learnable, and, if so, what determines their sample complexity?

We begin the chapter by showing that infinite classes can indeed be learnable, and thus, finiteness of the hypothesis class is not a necessary condition for learnability. We then present a remarkably crisp characterization of the family of learnable classes in the setup of binary valued classification with the zero-one loss. This characterization was first discovered by Vladimir Vapnik and Alexey Chervonenkis in 1970 and relies on a combinatorial notion called the Vapnik-Chervonenkis dimension (VC-dimension). We formally define the VC-dimension, provide several examples, and then state the fundamental theorem of statistical learning theory, which integrates the concepts of learnability, VC-dimension, the ERM rule, and uniform convergence.

6.1 INFINITE-SIZE CLASSES CAN BE LEARNABLE

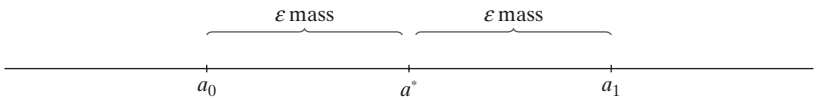
In Chapter 4 we saw that finite classes are learnable, and in fact the sample complexity of a hypothesis class is upper bounded by the log of its size. To show that the size of the hypothesis class is not the right characterization of its sample complexity, we first present a simple example of an infinite-size hypothesis class that is learnable.

Example 6.1. Let $\mathcal{H}$ be the set of threshold functions over the real line, namely, $\mathcal{H} = \{h_a : a \in \mathbb{R}\}$, where $h_a : \mathbb{R} \to \{0, 1\}$ is a function such that $h_a(x) = \mathbb{1}_{[x<a]}$. To remind the reader, $\mathbb{1}_{[x<a]}$ is 1 if $x < a$ and 0 otherwise. Clearly, $\mathcal{H}$ is of infinite size. Nevertheless, the following lemma shows that $\mathcal{H}$ is learnable in the PAC model using the ERM algorithm.

Lemma 6.1. *Let $\mathcal{H}$ be the class of thresholds as defined earlier. Then, $\mathcal{H}$ is PAC learnable, using the ERM rule, with sample complexity of $m_{\mathcal{H}}(\epsilon, \delta) \leq \lceil \log(2/\delta)/\epsilon \rceil$.*

Proof. Let $a^\star$ be a threshold such that the hypothesis $h^\star(x) = \mathbb{1}_{[x<a^\star]}$ achieves $L_{\mathcal{D}}(h^\star) = 0$. Let $\mathcal{D}_x$ be the marginal distribution over the domain $\mathcal{X}$ and let $a_0 < a^\star < a_1$ be such that

$$\mathbb{P}_{x\sim\mathcal{D}_x}[x \in (a_0, a^\star)] = \mathbb{P}_{x\sim\mathcal{D}_x}[x \in (a^\star, a_1)] = \epsilon.$$



(If $\mathcal{D}_x(-\infty, a^\star) \leq \epsilon$ we set $a_0 = -\infty$ and similarly for a_1). Given a training set S, let $b_0 = \max\{x : (x, 1) \in S\}$ and $b_1 = \min\{x : (x, 0) \in S\}$ (if no example in S is positive we set $b_0 = -\infty$ and if no example in S is negative we set $b_1 = \infty$). Let b_S be a threshold corresponding to an ERM hypothesis, h_S, which implies that $b_S \in (b_0, b_1)$. Therefore, a sufficient condition for $L_{\mathcal{D}}(h_S) \leq \epsilon$ is that both $b_0 \geq a_0$ and $b_1 \leq a_1$. In other words,

$$\mathbb{P}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(h_S) > \epsilon] \leq \mathbb{P}_{S\sim\mathcal{D}^m}[b_0 < a_0 \vee b_1 > a_1],$$

and using the union bound we can bound the preceding by

$$\mathbb{P}_{S\sim\mathcal{D}^m}[L_{\mathcal{D}}(h_S) > \epsilon] \leq \mathbb{P}_{S\sim\mathcal{D}^m}[b_0 < a_0] + \mathbb{P}_{S\sim\mathcal{D}^m}[b_1 > a_1]. \tag{6.1}$$

The event $b_0 < a_0$ happens if and only if all examples in S are not in the interval (a_0, a^*), whose probability mass is defined to be ϵ, namely,

$$\mathbb{P}_{S\sim\mathcal{D}^m}[b_0 < a_0] = \mathbb{P}_{S\sim\mathcal{D}^m}[\forall (x, y) \in S,\ x \notin (a_0, a^\star)] = (1-\epsilon)^m \leq e^{-\epsilon m}.$$

Since we assume $m > \log(2/\delta)/\epsilon$ it follows that the equation is at most $\delta/2$. In the same way it is easy to see that $\mathbb{P}_{S\sim\mathcal{D}^m}[b_1 > a_1] \leq \delta/2$. Combining with Equation (6.1) we conclude our proof. □

6.2 THE VC-DIMENSION

We see, therefore, that while finiteness of $\mathcal{H}$ is a sufficient condition for learnability, it is not a necessary condition. As we will show, a property called the VC-dimension of a hypothesis class gives the correct characterization of its learnability. To motivate the definition of the VC-dimension, let us recall the No-Free-Lunch theorem (Theorem 5.1) and its proof. There, we have shown that without restricting the hypothesis class, for any learning algorithm, an adversary can construct a distribution for which the learning algorithm will perform poorly, while there is another

learning algorithm that will succeed on the same distribution. To do so, the adversary used a finite set $C \subset \mathcal{X}$ and considered a family of distributions that are concentrated on elements of C. Each distribution was derived from a "true" target function from C to $\{0,1\}$. To make any algorithm fail, the adversary used the power of choosing a target function from the set of *all* possible functions from C to $\{0,1\}$.

When considering PAC learnability of a hypothesis class $\mathcal{H}$, the adversary is restricted to constructing distributions for which some hypothesis $h \in \mathcal{H}$ achieves a zero risk. Since we are considering distributions that are concentrated on elements of C, we should study how $\mathcal{H}$ behaves on C, which leads to the following definition.

Definition 6.2 (Restriction of $\mathcal{H}$ to C). Let $\mathcal{H}$ be a class of functions from $\mathcal{X}$ to $\{0,1\}$ and let $C = \{c_1, \ldots, c_m\} \subset \mathcal{X}$. The restriction of $\mathcal{H}$ to C is the set of functions from C to $\{0,1\}$ that can be derived from $\mathcal{H}$. That is,

$$\mathcal{H}_C = \{(h(c_1), \ldots, h(c_m)) : h \in \mathcal{H}\},$$

where we represent each function from C to $\{0,1\}$ as a vector in $\{0,1\}^{|C|}$.

If the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0,1\}$, then we say that $\mathcal{H}$ *shatters* the set C. Formally:

Definition 6.3 (Shattering). A hypothesis class $\mathcal{H}$ shatters a finite set $C \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0,1\}$. That is, $|\mathcal{H}_C| = 2^{|C|}$.

Example 6.2. Let $\mathcal{H}$ be the class of threshold functions over $\mathbb{R}$. Take a set $C = \{c_1\}$. Now, if we take $a = c_1 + 1$, then we have $h_a(c_1) = 1$, and if we take $a = c_1 - 1$, then we have $h_a(c_1) = 0$. Therefore, $\mathcal{H}_C$ is the set of all functions from C to $\{0,1\}$, and $\mathcal{H}$ shatters C. Now take a set $C = \{c_1, c_2\}$, where $c_1 \leq c_2$. No $h \in \mathcal{H}$ can account for the labeling $(0,1)$, because any threshold that assigns the label 0 to c_1 must assign the label 0 to c_2 as well. Therefore not all functions from C to $\{0,1\}$ are included in $\mathcal{H}_C$; hence C is not shattered by $\mathcal{H}$.

Getting back to the construction of an adversarial distribution as in the proof of the No-Free-Lunch theorem (Theorem 5.1), we see that whenever some set C is shattered by $\mathcal{H}$, the adversary is not restricted by $\mathcal{H}$, as they can construct a distribution over C based on *any* target function from C to $\{0,1\}$, while still maintaining the realizability assumption. This immediately yields:

Corollary 6.4. *Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0,1\}$. Let m be a training set size. Assume that there exists a set $C \subset \mathcal{X}$ of size $2m$ that is shattered by $\mathcal{H}$. Then, for any learning algorithm, A, there exist a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$ and a predictor $h \in \mathcal{H}$ such that $L_{\mathcal{D}}(h) = 0$ but with probability of at least $1/7$ over the choice of $S \sim \mathcal{D}^m$ we have that $L_{\mathcal{D}}(A(S)) \geq 1/8$.*

Corollary 6.4 tells us that if $\mathcal{H}$ shatters some set C of size $2m$ then we cannot learn $\mathcal{H}$ using m examples. Intuitively, if a set C is shattered by $\mathcal{H}$, and we receive a sample containing half the instances of C, the labels of these instances give us no information about the labels of the rest of the instances in C – every possible labeling of the rest of the instances can be explained by some hypothesis in $\mathcal{H}$. Philosophically,

If someone can explain every phenomenon, his explanations are worthless.

This leads us directly to the definition of the VC dimension.

Definition 6.5 (VC-dimension). The VC-dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the maximal size of a set $C \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size we say that $\mathcal{H}$ has infinite VC-dimension.

A direct consequence of Corollary 6.4 is therefore:

Theorem 6.6. *Let $\mathcal{H}$ be a class of infinite VC-dimension. Then, $\mathcal{H}$ is not PAC learnable.*

Proof. Since $\mathcal{H}$ has an infinite VC-dimension, for any training set size m, there exists a shattered set of size $2m$, and the claim follows by Corollary 6.4. □

We shall see later in this chapter that the converse is also true: A finite VC-dimension guarantees learnability. Hence, the VC-dimension characterizes PAC learnability. But before delving into more theory, we first show several examples.

6.3 EXAMPLES

In this section we calculate the VC-dimension of several hypothesis classes. To show that $\text{VCdim}(\mathcal{H}) = d$ we need to show that

1. There exists a set C of size d that is shattered by $\mathcal{H}$.
2. Every set C of size $d+1$ is not shattered by $\mathcal{H}$.

6.3.1 Threshold Functions

Let $\mathcal{H}$ be the class of threshold functions over $\mathbb{R}$. Recall Example 6.2, where we have shown that for an arbitrary set $C = \{c_1\}$, $\mathcal{H}$ shatters C; therefore $\text{VCdim}(\mathcal{H}) \geq 1$. We have also shown that for an arbitrary set $C = \{c_1, c_2\}$ where $c_1 \leq c_2$, $\mathcal{H}$ does not shatter C. We therefore conclude that $\text{VCdim}(\mathcal{H}) = 1$.

6.3.2 Intervals

Let $\mathcal{H}$ be the class of intervals over $\mathbb{R}$, namely, $\mathcal{H} = \{h_{a,b} : a, b \in \mathbb{R}, a < b\}$, where $h_{a,b} : \mathbb{R} \to \{0,1\}$ is a function such that $h_{a,b}(x) = \mathbb{1}_{[x \in (a,b)]}$. Take the set $C = \{1,2\}$. Then, $\mathcal{H}$ shatters C (make sure you understand why) and therefore $\text{VCdim}(\mathcal{H}) \geq 2$. Now take an arbitrary set $C = \{c_1, c_2, c_3\}$ and assume without loss of generality that $c_1 \leq c_2 \leq c_3$. Then, the labeling $(1,0,1)$ cannot be obtained by an interval and therefore $\mathcal{H}$ does not shatter C. We therefore conclude that $\text{VCdim}(\mathcal{H}) = 2$.

6.3.3 Axis Aligned Rectangles

Let $\mathcal{H}$ be the class of axis aligned rectangles, formally:

$$\mathcal{H} = \{h_{(a_1,a_2,b_1,b_2)} : a_1 \leq a_2 \text{ and } b_1 \leq b_2\}$$



Figure 6.1. Left: 4 points that are shattered by axis aligned rectangles. Right: Any axis aligned rectangle cannot label c_5 by 0 and the rest of the points by 1.

where

$$h_{(a_1,a_2,b_1,b_2)}(x_1,x_2) = \begin{cases} 1 & \text{if } a_1 \le x_1 \le a_2 \quad \text{and} \quad b_1 \le x_2 \le b_2 \\ 0 & \text{otherwise} \end{cases} \tag{6.2}$$

We shall show in the following that $\text{VCdim}(\mathcal{H}) = 4$. To prove this we need to find a set of 4 points that are shattered by $\mathcal{H}$, and show that no set of 5 points can be shattered by $\mathcal{H}$. Finding a set of 4 points that are shattered is easy (see Figure 6.1). Now, consider any set $C \subset \mathbb{R}^2$ of 5 points. In C, take a leftmost point (whose first coordinate is the smallest in C), a rightmost point (first coordinate is the largest), a lowest point (second coordinate is the smallest), and a highest point (second coordinate is the largest). Without loss of generality, denote $C = \{c_1, \ldots, c_5\}$ and let c_5 be the point that was not selected. Now, define the labeling $(1, 1, 1, 1, 0)$. It is impossible to obtain this labeling by an axis aligned rectangle. Indeed, such a rectangle must contain $c_1, \ldots, c_4$; but in this case the rectangle contains c_5 as well, because its coordinates are within the intervals defined by the selected points. So, C is not shattered by H, and therefore $\text{VCdim}(H) = 4$.

6.3.4 Finite Classes

Let $\mathcal{H}$ be a finite class. Then, clearly, for any set C we have $|\mathcal{H}_C| \le |\mathcal{H}|$ and thus C cannot be shattered if $|\mathcal{H}| < 2^{|C|}$. This implies that $\text{VCdim}(\mathcal{H}) \le \log_2(|\mathcal{H}|)$. This shows that the PAC learnability of finite classes follows from the more general statement of PAC learnability of classes with finite VC-dimension, which we shall see in the next section. Note, however, that the VC-dimension of a finite class $\mathcal{H}$ can be significantly smaller than $\log_2(|\mathcal{H}|)$. For example, let $\mathcal{X} = \{1, \ldots, k\}$, for some integer k, and consider the class of threshold functions (as defined in Example 6.2). Then, $|\mathcal{H}| = k$ but $\text{VCdim}(\mathcal{H}) = 1$. Since k can be arbitrarily large, the gap between $\log_2(|\mathcal{H}|)$ and $\text{VCdim}(\mathcal{H})$ can be arbitrarily large.

6.3.5 VC-Dimension and the Number of Parameters

In the previous examples, the VC-dimension happened to equal the number of parameters defining the hypothesis class. While this is often the case, it is not always true. Consider, for example, the domain $\mathcal{X} = \mathbb{R}$, and the hypothesis class $\mathcal{H} = \{h_\theta : \theta \in \mathbb{R}\}$ where $h_\theta : \mathcal{X} \to \{0, 1\}$ is defined by $h_\theta(x) = \lceil 0.5 \sin(\theta x) \rceil$. It is possible to prove that $\text{VCdim}(\mathcal{H}) = \infty$, namely, for *every* d, one can find d points that are shattered by $\mathcal{H}$ (see Exercise 6.8).

6.4 THE FUNDAMENTAL THEOREM OF PAC LEARNING

We have already shown that a class of infinite VC-dimension is not learnable. The converse statement is also true, leading to the fundamental theorem of statistical learning theory:

Theorem 6.7 (The Fundamental Theorem of Statistical Learning). *Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the $0-1$ loss. Then, the following are equivalent:*

1. *$\mathcal{H}$ has the uniform convergence property.*
2. *Any ERM rule is a successful agnostic PAC learner for $\mathcal{H}$.*
3. *$\mathcal{H}$ is agnostic PAC learnable.*
4. *$\mathcal{H}$ is PAC learnable.*
5. *Any ERM rule is a successful PAC learner for $\mathcal{H}$.*
6. *$\mathcal{H}$ has a finite VC-dimension.*

The proof of the theorem is given in the next section.

Not only does the VC-dimension characterize PAC learnability; it even determines the sample complexity.

Theorem 6.8 (The Fundamental Theorem of Statistical Learning – Quantitative Version). *Let $\mathcal{H}$ be a hypothesis class of functions from a domain $\mathcal{X}$ to $\{0,1\}$ and let the loss function be the $0-1$ loss. Assume that $\text{VCdim}(\mathcal{H}) = d < \infty$. Then, there are absolute constants C_1, C_2 such that*

1. *$\mathcal{H}$ has the uniform convergence property with sample complexity*

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}^{UC}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}$$

2. *$\mathcal{H}$ is agnostic PAC learnable with sample complexity*

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}$$

3. *$\mathcal{H}$ is PAC learnable with sample complexity*

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_{\mathcal{H}}(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}$$

The proof of this theorem is given in Chapter 28.

Remark 6.3. We stated the fundamental theorem for binary classification tasks. A similar result holds for some other learning problems such as regression with the absolute loss or the squared loss. However, the theorem does not hold for all learning tasks. In particular, learnability is sometimes possible even though the uniform convergence property does not hold (we will see an example in Chapter 13, Exercise 6.2). Furthermore, in some situations, the ERM rule fails but learnability is possible with other learning rules.

6.5 PROOF OF THEOREM 6.7

We have already seen that $1 \to 2$ in Chapter 4. The implications $2 \to 3$ and $3 \to 4$ are trivial and so is $2 \to 5$. The implications $4 \to 6$ and $5 \to 6$ follow from the No-Free-Lunch theorem. The difficult part is to show that $6 \to 1$. The proof is based on two main claims:

- If $\text{VCdim}(\mathcal{H}) = d$, then even though $\mathcal{H}$ might be infinite, when restricting it to a finite set $C \subset \mathcal{X}$, its "effective" size, $|\mathcal{H}_C|$, is only $O(|C|^d)$. That is, the size of $\mathcal{H}_C$ grows polynomially rather than exponentially with $|C|$. This claim is often referred to as *Sauer's lemma*, but it has also been stated and proved independently by Shelah and by Perles. The formal statement is given in Section 6.5.1 later.
- In Section 4 we have shown that finite hypothesis classes enjoy the uniform convergence property. In Section 6.5.2 later we generalize this result and show that uniform convergence holds whenever the hypothesis class has a "small effective size." By "small effective size" we mean classes for which $|\mathcal{H}_C|$ grows polynomially with $|C|$.

6.5.1 Sauer's Lemma and the Growth Function

We defined the notion of *shattering*, by considering the restriction of $\mathcal{H}$ to a finite set of instances. The growth function measures the maximal "effective" size of $\mathcal{H}$ on a set of m examples. Formally:

Definition 6.9 (Growth Function). Let $\mathcal{H}$ be a hypothesis class. Then the *growth function* of $\mathcal{H}$, denoted $\tau_{\mathcal{H}} : \mathbb{N} \to \mathbb{N}$, is defined as

$$\tau_{\mathcal{H}}(m) = \max_{C \subset \mathcal{X} : |C| = m} |\mathcal{H}_C|.$$

In words, $\tau_H(m)$ is the number of different functions from a set C of size m to $\{0, 1\}$ that can be obtained by restricting $\mathcal{H}$ to C.

Obviously, if $\text{VCdim}(\mathcal{H}) = d$ then for any $m \leq d$ we have $\tau_{\mathcal{H}}(m) = 2^m$. In such cases, $\mathcal{H}$ induces all possible functions from C to $\{0, 1\}$. The following beautiful lemma, proposed independently by Sauer, Shelah, and Perles, shows that when m becomes larger than the VC-dimension, the growth function increases polynomially rather than exponentially with m.

Lemma 6.10 (Sauer-Shelah-Perles). *Let $\mathcal{H}$ be a hypothesis class with* $\text{VCdim}(\mathcal{H}) \leq d < \infty$. *Then, for all m, $\tau_{\mathcal{H}}(m) \leq \sum_{i=0}^{d} \binom{m}{i}$. In particular, if $m > d + 1$ then $\tau_{\mathcal{H}}(m) \leq (em/d)^d$.*

Proof of Sauer's Lemma*

To prove the lemma it suffices to prove the following stronger claim: For any $C = \{c_1, \ldots, c_m\}$ we have

$$\forall \mathcal{H}, \quad |\mathcal{H}_C| \leq |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|. \tag{6.3}$$

The reason why Equation (6.3) is sufficient to prove the lemma is that if $\text{VCdim}(\mathcal{H}) \leq d$ then no set whose size is larger than d is shattered by $\mathcal{H}$ and therefore

$$|\{B \subseteq C : \mathcal{H} \text{ shatters } B\}| \leq \sum_{i=0}^{d} \binom{m}{i}.$$

When $m > d+1$ the right-hand side of the preceding is at most $(em/d)^d$ (see Lemma A.5 in Appendix A).

We are left with proving Equation (6.3) and we do it using an inductive argument. For $m = 1$, no matter what $\mathcal{H}$ is, either both sides of Equation (6.3) equal 1 or both sides equal 2 (the empty set is always considered to be shattered by $\mathcal{H}$). Assume Equation (6.3) holds for sets of size $k < m$ and let us prove it for sets of size m. Fix $\mathcal{H}$ and $C = \{c_1, \ldots, c_m\}$. Denote $C' = \{c_2, \ldots, c_m\}$ and in addition, define the following two sets:

$$Y_0 = \{(y_2, \ldots, y_m) : (0, y_2, \ldots, y_m) \in \mathcal{H}_C \vee (1, y_2, \ldots, y_m) \in \mathcal{H}_C\},$$

and

$$Y_1 = \{(y_2, \ldots, y_m) : (0, y_2, \ldots, y_m) \in \mathcal{H}_C \wedge (1, y_2, \ldots, y_m) \in \mathcal{H}_C\}.$$

It is easy to verify that $|\mathcal{H}_C| = |Y_0| + |Y_1|$. Additionally, since $Y_0 = \mathcal{H}_{C'}$, using the induction assumption (applied on $\mathcal{H}$ and C') we have that

$$|Y_0| = |\mathcal{H}_{C'}| \leq |\{B \subseteq C' : \mathcal{H} \text{ shatters } B\}| = |\{B \subseteq C : c_1 \notin B \wedge \mathcal{H} \text{ shatters } B\}|.$$

Next, define $\mathcal{H}' \subseteq \mathcal{H}$ to be

$$\begin{aligned}\mathcal{H}' &= \{h \in \mathcal{H} : \exists h' \in \mathcal{H} \text{ s.t. } (1 - h'(c_1), h'(c_2), \ldots, h'(c_m)) \\ &= (h(c_1), h(c_2), \ldots, h(c_m))\},\end{aligned}$$

namely, $\mathcal{H}'$ contains pairs of hypotheses that agree on C' and differ on c_1. Using this definition, it is clear that if $\mathcal{H}'$ shatters a set $B \subseteq C'$ then it also shatters the set $B \cup \{c_1\}$ and vice versa. Combining this with the fact that $Y_1 = \mathcal{H}'_{C'}$ and using the inductive assumption (now applied on $\mathcal{H}'$ and C') we obtain that

$$\begin{aligned}|Y_1| &= |\mathcal{H}'_{C'}| \leq |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B\}| = |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B \cup \{c_1\}\}| \\ &= |\{B \subseteq C : c_1 \in B \wedge \mathcal{H}' \text{ shatters } B\}| \leq |\{B \subseteq C : c_1 \in B \wedge \mathcal{H} \text{ shatters } B\}|.\end{aligned}$$

Overall, we have shown that

$$\begin{aligned}|\mathcal{H}_C| &= |Y_0| + |Y_1| \\ &\leq |\{B \subseteq C : c_1 \notin B \wedge \mathcal{H} \text{ shatters } B\}| + |\{B \subseteq C : c_1 \in B \wedge \mathcal{H} \text{ shatters } B\}| \\ &= |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|,\end{aligned}$$

which concludes our proof.

6.5.2 Uniform Convergence for Classes of Small Effective Size

In this section we prove that if $\mathcal{H}$ has small effective size then it enjoys the uniform convergence property. Formally,

3.5.5 Prove that there exists a constant $K < \infty$ such that if $\mathcal{F}$ is such that $\rho_1(f) \le \delta$, for all $f \in \mathcal{F}$, where ρ_1 is the Bernstein size, then

$$E\left\|\sum_{i=1}^{n}(f(X_i)-Pf)\right\|_{\mathcal{F}}^{*} \le K\left(1+\frac{1}{\delta^2}\int_0^{\delta}\sqrt{H_{B,1}(\mathcal{F},P,\varepsilon)}\,d\varepsilon\right)\int_0^{\delta}\sqrt{H_{B,1}(\mathcal{F},P,\varepsilon)}\,d\varepsilon.$$

Hint: Improve on some of the estimates in the proof of Theorem 3.5.21 along the lines of the proof of Proposition 3.5.15.

3.5.6 Modify the proof of Proposition 3.5.17 to show that if $\mathcal{F}$ is a collection of uniformly bounded monotone functions on $\mathbb{R}$, then uniformly in P, for all $p \ge 1$, $\log N_{[]}(\mathcal{F}, L^p(P), \delta) \le K/\delta$.

3.6 Vapnik-Červonenkis Classes of Sets and Functions

Many classes of sets and functions used in statistical applications of empirical processes have the remarkable property of admitting bounds for their $L^2(P)$ covering numbers $N(\mathcal{F}, L^2(P), \varepsilon)$ of the order of ε^{-s}, for some $s < \infty$, *uniformly* in $P \in \mathcal{P}(S)$, where $\mathcal{P}(S)$ is the set of all probability measures on $(S, \mathcal{S})$. Consider as a first example linearly ordered uniformly bounded classes of functions, as in the exercises from the preceding section. Any empirical process indexed by these classes admits excellent bounds for the expectation of its supremum, and these, combined with Talagrand's inequality, produce exponential bounds that are of Gaussian type in large portions of their range, that is, best possible at least up to constants. It will be seen in later sections that the usual limit theorems do hold uniformly in P for these classes. These classes of sets were discovered by Vapnik and Červonenkis and their entropy properties by Dudley who called them *VC classes*. The entropy properties of *VC* classes of sets are inherited by related classes of functions that go by the name of *VC subgraph classes* of functions. Other related classes of functions such as *VC*-major and *VC*-hull classes admit larger but still manageable uniform bounds. In this section, *VC* classes of sets and the related class functions are defined and their $L^2(P)$ metric entropy properties established.

3.6.1 Vapnik-Červonenkis Classes of Sets

Let $\mathcal{C}$ be a class of subsets of a set S. Let $A \subseteq S$ be a finite set. The *trace* of $\mathcal{C}$ on A is the collection of all the subsets of A obtained by intersection of A with sets C from $\mathcal{C}$. Denote by $\Delta^{\mathcal{C}}(A)$ the cardinality of the trace of the class $\mathcal{C}$ on A. Then $\Delta^{\mathcal{C}}(A) \le 2^{\mathrm{Card}(A)}$, and we say that $\mathcal{C}$ *shatters* A if $\Delta^{\mathcal{C}}(A) = 2^{\mathrm{Card}(A)}$, that is, if every subset of A is the intersection of A with some set $C \in \mathcal{C}$. Let

$$m^{\mathcal{C}}(k) = \sup_{\substack{A \subseteq S \\ \mathrm{Card}(A)=k}} \Delta^{\mathcal{C}}(A).$$

Definition 3.6.1 A collection of sets $\mathcal{C}$ is a Vapnik-Červonenkis class ($\mathcal{C}$ is *VC*) if the quantity

$$v(\mathcal{C}) := \begin{cases} \min\{k : m^{\mathcal{C}}(k) < 2^k\} & \text{if } m^{\mathcal{C}}(k) < 2^k \text{ for some } k < \infty \\ \infty & \text{otherwise} \end{cases}$$

is finite, that is, if there exists $k < \infty$ such that $\mathcal{C}$ does not shatter any subsets of S of cardinality k. The *VC* index of the class $\mathcal{C}$ is defined as $v(\mathcal{C})$.

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For instance, the collection of half-lines $\mathcal{C} = \{(-\infty, t] \ : \ t \in \mathbb{R}\}$ is *VC* and $v(\mathcal{C}) = 2$; if $x_1 < x_2$, then $\Delta^{\mathcal{C}}(x_1, x_2) = 3 < 2^2$ because $\mathcal{C}$ cannot pick out $\{x_2\}$.

It is easy to see that $v(\mathcal{C}) = 0$ if and only if $\mathcal{C}$ is empty and $v(\mathcal{C}) = 1$ if and only if $\mathcal{C}$ consists of only one set (Exercise 3.6.1). In the first case, $m^{\mathcal{C}}(n) = 0$ for all n, and in the second, $m^{\mathcal{C}}(n) = 1$ for all $n \geq 1$.

The following remarkable combinatorial theorem asserts that *either* $m^{\mathcal{C}}(k) = 2^k$ *for all* k *or* $m^{\mathcal{C}}(k)$ *grows only polynomially in* k. The main results on *VC* classes of sets and functions are based on this theorem.

Theorem 3.6.2 *Let* $\mathcal{C}$ *be a non-empty VC class, and let* $v = v(\mathcal{C}\Rightarrow$. *Then, for any finite set* $A \subseteq S$,

$$\Delta^{\mathcal{C}}(A) \leq \text{Card}\{B \subseteq A : \ \text{Card}(B) < v\} = \sum_{j=0}^{v-1} \binom{\text{Card}(A)}{j},$$

and therefore,

$$m^{\mathcal{C}}(n) \leq \sum_{j=0}^{v-1} \binom{n}{j},$$

a polynomial in n of degree $v - 1$. *In particular, there is a constant* $B(v) < \infty$ *depending only on v such that* $m^{\mathcal{C}}(n) \leq B(v) n^{v-1}$, *for all* $n \geq v - 1$.

The result follows from the following proposition by letting $\mathcal{U} = \{C \cap A : \ C \in \mathcal{C}\}$:

Proposition 3.6.3 *Let A be a finite set, and let* $\mathcal{U}$ *be a class of subsets of A. Then*

$$\text{Card}(\mathcal{U}) \leq \text{Card}\{B \subseteq A : \ B \text{ is shattered by } \mathcal{U}\}.$$

Proof Say that a class of subsets of A, $\mathcal{U}'$, is hereditary if when a set B is in $\mathcal{U}'$ then all the subsets of B are also in $\mathcal{U}'$. By definition, if $\mathcal{U}'$ is hereditary, then $\mathcal{U}'$ shatters all the sets it contains, and therefore,

$$\text{Card}(\mathcal{U}') \leq \text{Card}\{B \subseteq A : \ B \text{ is shattered by } \mathcal{U}'\}.$$

Hence, the proposition will be proved if we construct a collection $\mathcal{U}'$ from $\mathcal{U}$ such that

(i) $\mathcal{U}'$ is hereditary,
(ii) $\text{Card}(\mathcal{U}) = \text{Card}(\mathcal{U}')$, and
(iii) $\mathcal{U}$ shatters at least as many sets as $\mathcal{U}'$ shatters.

We will obtain $\mathcal{U}'$ by repeated application of a simple transformation T_x that with each application renders the class of sets a step closer to being hereditary. Given $x \in A$ and a collection $\mathcal{V}$ of subsets of A, let $T_x^{\mathcal{V}}(\mathcal{V}) := T_x(\mathcal{V}) := \{T_x(V) : \ V \in \mathcal{V}\}$, where, for $V \in \mathcal{V}$,

$$T_x(V) = \begin{cases} V \setminus \{x\} & \text{if } x \in V \text{ and } V \setminus \{x\} \notin \mathcal{V} \\ V & \text{otherwise.} \end{cases}$$

Thus all the sets $V \in T_x(\mathcal{V})$ satisfy that if $x \in V$ then $V \setminus \{x\} \in T_x(\mathcal{V})$, which is clearly a step towards transforming $\mathcal{V}$ into a hereditary set, as mentioned earlier. Another property of $T_x(\mathcal{V})$ to be used later that follows directly from the definition is that if $V \in T_x(\mathcal{V})$ and $x \in V$, then $V \in \mathcal{V}$, $V \setminus \{x\} \in \mathcal{V}$ and $T_x(V) = V$.

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Define, on classes $\mathcal{V}$ of subsets of A, the functional $w(\mathcal{V}) := \sum_{V\in\mathcal{V}} \mathrm{Card}(V)$. Then, since $\mathrm{Card}(T_x^{\mathcal{V}}(V))$ is either $\mathrm{Card}(V)$ or $\mathrm{Card}(V)-1$, it follows that $w \circ T_x \le w$. This implies that the minimum of w among all the collections $\mathcal{U}'$ obtained from $\mathcal{U}$ by finitely many repeated applications of the maps T_x, $x \in A$, is attained. Let $\mathcal{U}'$ be one of the possibly more than one such collections on which w is minimal. Then $w(\mathcal{U}') = w(T_x^{\mathcal{U}'}(\mathcal{U}'))$, for all $x \in A$. But this implies that $\mathcal{U}'$ is hereditary if $\mathcal{U}'$ were not hereditary, then there would exist a set $B \in \mathcal{U}'$ and a point $x \in A$ such that $B \setminus \{x\} \notin \mathcal{U}'$, implying that $\mathrm{Card}(T_x(B)) < \mathrm{Card}(B)$ and hence also that $w(\mathcal{U}') < w(T_x(\mathcal{U}')$, a contradiction. (Repeated applications of the maps T_x means, for example, that $T_x^{T_y\mathcal{U}}(T_y^{\mathcal{U}}(B))$.)

Since $\mathcal{U}'$ is obtained from $\mathcal{U}$ by a finite number of operations T_x, to prove that $\mathcal{U}'$ satisfies preceding properties (ii) and (iii), it suffices to see that they are satisfied by $T_x(\mathcal{U})$. Thus, we must show that $\mathrm{Card}(T_x(\mathcal{U})) = \mathrm{Card}(\mathcal{U})$ and that $\mathcal{U}$ shatters at least as many sets as $T_x(\mathcal{U})$ does. To prove the first property, we observe that T_x is one to one: the operation T_x keeps unchanged those sets $U \in \mathcal{U}$ which contain x and are such that $U \setminus \{x\}$ is in $\mathcal{U}$ as well as those sets $U \in \mathcal{U}$ which do not contain x, and T_x replaces U by $U \setminus \{x\}$ if U contains x but $U \setminus \{x\}$ is not in $\mathcal{U}$. This is obviously a one-to-one replacement. To see the second property, we must show that if $T_x(\mathcal{U})$ shatters B, then $\mathcal{U}$ also shatters B. Suppose that $T_x(\mathcal{U})$ shatters B. If $x \notin B$, then $\mathcal{U}$ shatters B because $U \in \mathcal{U}$ differs from $T_x(U) \in T_x(\mathcal{U})$ at most by x. If $x \in B$, then for every $B' \subseteq B\setminus\{x\}$ there exists $V \in T_x(\mathcal{U})$ such that $V \cap B = B' \cup \{x\}$ because $T_x(\mathcal{U})$ shatters B. This implies that $x \in V$ and therefore that V is also in $\mathcal{U}$, thus showing that $U \cap B = B' \cup \{x\}$ for some $U \in \mathcal{U}$; moreover, $V \setminus \{x\} \in \mathcal{U}$ and $(V \setminus \{x\}) \cap B = B'$. Hence, $\mathcal{U}$ shatters B. ■

The following proposition will simplify the bound for $m^{\mathcal{C}}(n)$ in the preceding theorem.

Proposition 3.6.4 *Let k and n be nonnegative integers such that $n \ge k+2$. Then*

$$\sum_{j=0}^{k} \binom{n}{j} \le \frac{1.5n^k}{k}. \tag{3.229}$$

Proof We recall the nonasymptotic Stirling's formula (see Feller, vol. I, 1968, p. 52)

$$(n/e)^n\sqrt{2\pi n} \le n \le e^{1/12n}(n/e)^n\sqrt{2\pi n}, \quad n \in \mathbb{N}, \tag{3.230}$$

to be used at a crucial step in this proof. We also recall the 'Pascal triangle' property of the quantities $C_{n,\le k} := \sum_{j=0}^{k} \binom{n}{j}$, simply inherited from the same property of the combinatorial numbers $\binom{n}{k}$

$$C_{n,\le k} = C_{n-1,\le k} + C_{n-1,\le k-1}, \quad k,n \in \mathbb{N},\ k < n. \tag{3.231}$$

Another inequality we need is the following simple one: for $n,k \ge 1$,

$$\frac{n^k}{k} \ge \frac{(n-1)^k}{k} + \frac{(n-1)^{k-1}}{(k-1)}, \tag{3.232}$$

which follows because, by the binomial theorem, $n^k \ge (n-1)^k + k(n-1)^{k-1}$. Suppose that the proposition is true for $C_{n,\le n-2}$ for all n and that, given K, the proposition is also true for $C_{n,\le k}$, for all (n,k) with $0 \le k \le K-1$ and $n \ge k+2$. Then, using (3.231) and (3.232),

$$C_{K+3,\le K} = C_{K+2,\le K} + C_{K+2,\le K-1} \le \frac{1.5(K+2)^K}{K} + \frac{1.5(K+2)^{K-1}}{(K-1)} \le \frac{1.5(K+3)^K}{K}.$$

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Consequently,

$$C_{K+4,\le K} = C_{K+3,\le K} + C_{K+3,\le K-1} \le \frac{1.5(K+4)^K}{K}$$

and, in general, $C_{n,\le K} \le 1.5n^K/K$, for all $n \ge K+2$. Hence, the proposition will follow by induction on k if we show that it holds for the pairs (n,k) such that $k=0$ and $n \ge 2$, $k=1$ and $n \ge 3$ and $(n,n-2)$ for all $n \ge 3$. For $k=0$, it follows because $C_{n,\le 0}=1$, and for $k=1$ and $n \ge 3$, it follows because $1+n \le 1.5n$. For $k=n-2$, we must show that

$$2^n - 1 - n \le \frac{1.5n^{n-2}}{(n-2)} = \frac{1.5(n-1)n^{n-1}}{n}.$$

This can be checked directly for $n=3,\ldots,6$. For $n \ge 7$, we will prove the slightly stronger inequality $2^n n \le 1.5(n-1)n^{n-1}$. By Stirling's formula, it suffices to show that

$$2^n(n/e)^n(2\pi n)^{1/2}e^{\frac{1}{12n}} \le 1.5(n-1)n^{n-1}.$$

Taking logarithms and then derivatives, it is easy to see that $(e/2)^n \ge 2n^{1/2}$ for all $n \ge 7$ (in fact, $n \ge 5$). But then the preceding inequality follows from

$$\sqrt{2\pi}\,e^{\frac{1}{12n}} \le 3(1-1/n),$$

which does hold for all $n \ge 7$ ($\sqrt{2\pi} \le 2.51$, $18/7 \ge 2.57$ and the exponential is very close to 1). ■

The preceding proposition and theorem give the following:

Corollary 3.6.5 *If $\mathcal{C}$ is a non-empty VC class of sets and $v = v(\mathcal{C})$ is its VC index, then*

$$m^{\mathcal{C}}(n) \le \frac{1.5n^{v-1}}{(v-1)}, \quad \text{for } n \ge v+1.$$

For $n = v$, $m^{\mathcal{C}}(n) \le 2^v - 1$, and for $n < v$, $m^{\mathcal{C}}(n) \le 2^n < 2^v - 1$. In particular,

$$m^{\mathcal{C}}(n) \le 2n^{v-1}, \quad \text{for all } n \ge 1.$$

Whereas the inequalities in this corollary are not sharp, the inequality in Theorem 3.6.2 is (see Exercise 3.6.2). The preceding inequality follows easily for $v(\mathcal{C}) \ge 2$, and it follows from Exercise 3.6.1 for $v=0$ and $v=1$ (in these cases, $m^{\mathcal{C}}(n)$ is, respectively, 0 and 1).

It is not always easy to prove that a class of sets is *VC*. The following two propositions can be very helpful:

Proposition 3.6.6 *If $\mathcal{G}$ is a finite-dimensional vector space of real functions on S, then the class of sets $\mathcal{C} := [\{g \ge 0\} : g \in \mathcal{G}\uplus$ is VC with $v(\mathcal{C}) = \dim\mathcal{G} + 1$. The same is true for $[\{g > 0\} : g \in \mathcal{G}]$.*

Proof Let $v - 1 = \dim\mathcal{G}$, and let $\{s_1,\ldots,s_v\}$ be v distinct points of S. (The result clearly holds if S contains fewer than v points.) Let $L : \mathcal{G} \mapsto \mathbb{R}^v$ be given by $L(g) = (g(s_1),\ldots,g(s_v))$. Then $\dim L(\mathcal{G}) \le v-1$. Let $w = (w_1,\ldots,w_v)$ be a nonzero vector orthogonal to $L(\mathcal{G})$. Then

$$\sum w_i I(w_i \ge 0) g(s_i) = -\sum w_i I(w_i < 0) g(s_i),$$

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and we can assume that the set of i with $w_i < 0$ is not empty (otherwise, we can replace w by $-w$). If there existed g for which $\{g \geq 0\} \cap \{s_1, \ldots, s_v\} = \{s_i : w_i \geq 0\}$, then the left side of the preceding equation for this g would be larger than or equal to zero, whereas the right side would be negative, which is impossible. Thus, there is a subset of $\{s_1, \ldots, s_v\}$ that is not the intersection of $\{s_1, \ldots, s_v\}$ with any set $\{g \geq 0\}$. Hence, the class $\mathcal{C}$ is *VC*, and $v(\mathcal{C}) \leq \dim \mathcal{G} + 1$.

Now let $m = v - 1 = \dim(G)$. Consider the vector space $G'_S = \{\sum a_i \delta_{x_i} : a_i \in \mathbb{R},\ x_i \in S\}$, where δ_x is unit mass at x, and note that the natural inclusion $G \subset (G'_S)'$ is injective (if two functions are different, they are also different as linear functionals on G'_S). Hence, the dimension of G'_S, which equals that of its dual, is at least m. Since $\{\delta_x : x \in S\}$ generates G'_S, it follows that there exist m points $x_i \in S$ such that $\delta_{x_1}, \ldots, \delta_{x_m}$ are linearly independent as elements of G'_S. Therefore, $\{(g(x_1), \ldots, g(x_m)) : g \in G\} = \mathbb{R}^m$, which implies that every subset of the set $\{x_1, \ldots, x_m\}$ is the intersection of this set with $\{g \geq 0\}$ for some $g \in G$. Thus, $v(\mathcal{C}) \geq m + 1 = v$. The second assertion follows in a similar way. ■

For example, this proposition immediately gives that the class of all closed half-spaces of $\mathbb{R}^d$ is *VC* and so is the class of all the open half-spaces. The same is true for the class of all the closed balls (or all open balls) of $\mathbb{R}^d$.

Exercise 3.6.4 and the following proposition are examples of permanence properties of *VC* classes.

Proposition 3.6.7

(i) *If $\mathcal{C}$ is VC, then $\mathcal{C}^c := \{C^c : C \in \mathcal{C}\}$ is VC.*
(ii) *If $\mathcal{C}$ and $\mathcal{D}$ are VC, then $\mathcal{C} \cup \mathcal{D}$ and $\mathcal{C} \cap \mathcal{D}$ are VC. Here $\mathcal{C} \cup \mathcal{D} = \{C \cup D :\ C \in \mathcal{C}, D \in \mathcal{D}\}$ and $\mathcal{C} \cap \mathcal{D} = \{C \cap D :\ C \in \mathcal{C}, D \in \mathcal{D}\}$.*
(iii) *If $\mathcal{C}$ is a collection of subsets of S and $\mathcal{D}$ is a collections of subsets of T and both are VC, then $\mathcal{C} \times \mathcal{D}$ is also VC, where $\mathcal{C} \times \mathcal{D} = \{C \times D :\ C \in \mathcal{C}, D \in \mathcal{D}\}$.*
(iv) *If $\mathcal{C} \subset \mathcal{D}$ and $\mathcal{D}$ are VC, then $\mathcal{C}$ is VC.*

Proof Obviously, $\mathcal{C}$ shatters A if and only if $\mathcal{C}^c$ does, and we also have $v(\mathcal{C}) = v(\mathcal{C}^c)$. The trace of $\mathcal{C}$ on a set of n points consists of a number of sets not exceeding $1.5n^{v(\mathcal{C})-1}$, and the trace of $\mathcal{D}$ on each of these subsets does not exceed $1.5n^{v(\mathcal{D})-1}$; hence, the trace of $\mathcal{C} \cap \mathcal{D}$ does not exceed $2.25n^{v(\mathcal{C})+v(\mathcal{D})-2}$, smaller than 2^n for n large enough. This also proves $\mathcal{C} \cup \mathcal{D}$ is *VC* by taking complements and applying the first property. If $A \subset S \times T$ has n points, then clearly $\Delta^{\mathcal{C}\times T}(A) = \Delta^{\mathcal{C}}(\pi_1(A))$, where $\pi_1(s,t) = s$, and likewise for $S \times \mathcal{D}$, showing that these two classes are *VC*; hence, so is their intersection $\mathcal{C} \times \mathcal{D}$. ■

The preceding two propositions show, for example, that the class of all polygons of $\mathbb{R}^2$ of less than k sides, any fixed k, is *VC*, and likewise in $\mathbb{R}^d$ for piecewise polynomial regions of a fixed finite number of pieces and degrees not exceeding a fixed number k.

We conclude this subsection by showing in the most naive way possible how Theorem 3.6.2 can be used to evaluate empirical processes indexed by *VC* classes of sets. The reader will find it meaningful because we already have a relatively large collection of examples of *VC* classes. Let $\mathcal{C}$ be a countable, not empty *VC* class of subsets of S of index v, and consider the empirical processes indexed by $\mathcal{C}$ and based on an i.i.d. sample $\{X_i\}$ from

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P. Let

$$\tilde{P}_n(C) = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i \delta_{X_i}(C) = \frac{1}{n}\sum_{i=1}^{n} \varepsilon_i I_{X_i \in C}, \quad C \in \mathcal{C},$$

be the empirical process randomised by an independent Rademacher sequence $\{\varepsilon_i\}$ independent of the variables X_j. Observe that if $\{X_1,\ldots,X_n\} \cap C = \{X_1,\ldots,X_n\} \cap D$, then $\tilde{P}_n(C) = \tilde{P}_n(D)$. This implies by Corollary 3.6.5 that $\sup_{C\in\mathcal{C}} |\tilde{P}_n(C)|$ is in fact the maximum over at most $m^{\mathcal{C}}(n) \le 1.5n^{v-1}/(v-1)$ random variables of the form $n^{-1}\sum_{i=1}^{n} \varepsilon_i a_i$, with $a_i = 0$ or $a_i = 1$. But these random variables are sub-Gaussian with $\sigma^2 \le 1/n$. Hence, the simple maximal inequality for finite collections of sub-Gaussian variables in Lemma 2.3.4 together with randomisation yields, for $n \ge 2$,

$$E\|P_n - P\|_{\mathcal{C}} \le 2E\|\tilde{P}_n\|_{\mathcal{C}} = 2E_X E_\varepsilon \|\tilde{P}_n\|_{\mathcal{C}} \le 2\sqrt{2\log\frac{1.5n^{v-1}}{(v-1)}}/\sqrt{n} \to 0,$$

a uniform Glivenko-Cantelli theorem for these classes. The first identity here requires measurability, and this is why we are assuming $\mathcal{C}$ to be countable, an assumption that can be relaxed but not altogether ignored. This is the celebrated Vapnik-Červonenkis law of large numbers, which does go far beyond the multivariate Glivenko-Cantelli theorem. Strictly speaking, the Glivenko-Cantelli theorem states a.s. convergence, but here, once convergence to zero in probability is established, one simply uses the reverse submartingale convergence theorem to show convergence a.s. (Exercise 3.6.8).

For more sophisticated applications, a bound on the $L^2(P)$-covering numbers of VC classes of sets is needed. This is given in the next subsection, directly for VC subgraph classes of functions, which include in particular VC classes of sets.

3.6.2 VC Subgraph Classes of Functions

In this subsection we consider classes of functions on S that are related in different ways to VC classes of sets. The most interesting are probably the VC subgraph classes. The main result is Theorem 3.6.9, which shows that the $L^p(P)$-covering numbers of these classes of functions admit small bounds, of the order of $\varepsilon^{-(v-1)p}$, uniformly in P.

Definition 3.6.8 The subgraph of a real function f on S is the set

$$G_f = \{(s,t) :\ s \in S,\ t \in \mathbb{R},\ t \le f(s)\}.$$

A class of functions $\mathcal{F}$ is *VC subgraph* if the class of sets $\mathcal{C} = \{G_f :\ f \in \mathcal{F}\}$ is VC.

The family of indicator functions of the sets in a VC class is a VC subgraph class of functions. More generally, if f is a function on S and $\mathcal{C}$ is a VC class of sets, then the class of functions $\mathcal{F} = \{fI_C : C \in \mathcal{C}\}$ is VC subgraph. To see this, we just note that for a subset $\{(s_1,t_1),\ldots,(s_n,t_n)\}$ of $S \times \mathbb{R}$ to be shattered by the subgraphs of functions in $\mathcal{F}$, it is necessary that the S coordinates $s_1,\ldots,s_n$ be all different and that the set $\{s_1,\ldots,s_n\}$ be shattered by $\mathcal{C}$.

Hence, the following key proposition for VC subgraph classes of functions applies to VC classes of sets and beyond. First, here is some notation. Let $\mathcal{F}$ be a class of functions in

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$L^p(S,\mathcal{S},P)$, $0<p<\infty$. Then, letting $e_{p,P}=e_p$ denote the $L^p(P)$ pseudo-distance $e_p(f,g)=\|f-g\|_p=\int|f-g|^p dP$ for $0<p<1$, $e_p(f,g)=\|f-g\|_p=\left[\int|f-g|^p dP\right]^{1/p}$ for $p\geq 1$, we write, as usual,

$$D(\mathcal{F},L^p(P),\varepsilon):=D(\mathcal{F},e_p,\varepsilon)$$

and likewise for N_p. We recall that given a class of functions $\mathcal{F}$, a *measurable envelope* or an *envelope* F of $\mathcal{F}$ is any *everywhere finite, measurable* function F such that

$$\sup_{f\in\mathcal{F}}|f(s)|\leq F(s),\quad s\in S.$$

Of course, a class $\mathcal{F}$ may not admit any measurable envelopes. Finally, we should emphasise that although $\mathcal{F}$ is always a subset of measurable functions on a probability space, we do not identify functions which are a.s. equal.

Theorem 3.6.9 (**Dudley-Pollard**) *Let $\mathcal{F}$ be a non-empty VC subgraph class of functions admitting an envelope $F\in L^p(S,\mathcal{S},P)$ for some $0<p<\infty$ which is assumed, without loss of generality, bounded away from zero. Suppose that the class $\mathcal{C}$ of subgraphs of the functions in $\mathcal{F}$ has index v. Set $m_{v,w}=\max\{m\in\mathbb{N}:\log m\geq m^{1/(v-1)-1/w}\}$ for $w>v-1$. Then we have*

$$D(\mathcal{F},L^p(P),\varepsilon\|F\|_p)\leq m_{v,w}\vee\left[2^{w/(v-1)}\left(\frac{2^{p+1}}{\varepsilon^p}\right)^w\right],\quad \textit{for all } w>v-1 \textit{ if } p\geq 1, \tag{3.233}$$

and

$$D(\mathcal{F},L^p(P),\varepsilon\|F\|_p)\leq m_{v,w}\vee\left[2^{w/(v-1)}\left(\frac{2}{\varepsilon^{1/p}}\right)^w\right],\quad \textit{for all } w>v-1 \textit{ if } p<1. \tag{3.234}$$

Hence, the same bounds apply to $N(\mathcal{F},L^p(P),\varepsilon\|F\|_p)$.

Proof The theorem is proved if $D(\mathcal{F},L^p(P),\varepsilon\|F\|_p)\leq 1$, so we may assume the packing numbers of the class $\mathcal{C}$ of subgraphs to be larger than 1 and, in particular, $v\geq 2$. First, we consider the case $p\geq 1$. Let $f_1,\ldots,f_m$ be a maximal collection of functions in $\mathcal{F}$ for which

$$P|f_i-f_j|^p>\varepsilon^p PF^p,\quad i\neq j,$$

so that $m=D(\mathcal{F},L^p(P),\varepsilon\|F\|_p)$. For k, to be specified later, let (s_i,t_i), $1\leq i\leq k$, be independent identically distributed $(S\times\mathbb{R})$ random vectors with law

$$\Pr\{(s,t)\in A\times[a,b]\}=\frac{\int_A\lambda[(-F(s))\vee a,F(s)\wedge b]F^{p-1}(s)dP(s)}{2PF^p},$$

for $A\in\mathcal{S}$ and real numbers $a<b$, where λ is Lebesgue measure. (That is, independently for each i, s_i is chosen according to the law $P_F(A)=P(I_AF^p)/PF^p$, $A\in\mathcal{S}$, and given s_i, t_i is chosen uniformly on $[-F(s_i),F(s_i)]$.) Let C_i denote the subgraph of f_i. The probability that

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at least two graphs have the same intersection with the sample $\{(s_i, t_i),\ i \le k\}$ is at most

$$\begin{aligned}
&\binom{m}{2} \max_{i\neq j} \Pr\{C_i \text{ and } C_j \text{ have the same intersection with the sample}\} \\
&= \binom{m}{2} \max_{i\neq j} \prod_{r=1}^{k} \Pr\{(s_r,t_r) \notin C_i \Delta C_j\} \\
&= \binom{m}{2} \max_{i\neq j} \prod_{r=1}^{k} \Big[1 - \Pr\{(s_r,t_r) \in C_i \Delta C_j\}\Big] \\
&= \binom{m}{2} \max_{i\neq j} \prod_{r=1}^{k} \Big[1 - \Pr\big\{(s_r,t_r) : t_r \text{ is between } f_i(s_r), f_j(s_r)\big\}\Big] \\
&= \binom{m}{2} \max_{i\neq j} \left[1 - \frac{1}{\|F\|_p^p} \int \frac{|f_i - f_j|}{2F} F^p dP\right]^k \\
&\le \binom{m}{2} \max_{i\neq j} \left[1 - \frac{1}{\|F\|_p^p} \int \frac{|f_i - f_j|^p}{2^p} dP\right]^k \\
&\le \binom{m}{2} \left[1 - \frac{\varepsilon^p}{2^p}\right]^k \\
&\le \binom{m}{2} \exp\left(-\frac{\varepsilon^p k}{2^p}\right).
\end{aligned}$$

Let k be such that this probability is strictly less than 1. Then there exists a set of k elements of S such that graphs $C_i \in \mathcal{C}$, $1 \le i \le m$, intersect different subsets of this set, which implies that $m^{\mathcal{C}}(k) \ge m$. Since $2^p(\log 2)/\varepsilon^p > 1$ for $p \ge 1$ and $0 < \varepsilon \le 1$, the smallest integer k such that $\binom{m}{2} \exp\left(-\frac{\varepsilon^p k}{2^p}\right) < 1$ satisfies $1 \le k \le (2^{p+1}/\varepsilon^p) \log m$. Therefore, by Corollary 3.6.5, we have

$$m \le m^{\mathcal{G}}(k) \le 2k^{v-1} \le 2\left(\frac{2^{p+1}}{\varepsilon^p} \log m\right)^{v-1}.$$

Then, given $w > v - 1$ and setting $\tau = 1/(v-1) - 1/w$, either $m \le m_{v,w}$ or $m \le 2\left((2^p/\varepsilon^p) m^\tau\right)^{v-1}$, proving (3.233).

The proof for $p < 1$ is similar, with the following changes: the functions f_i satisfy $P|f_i - f_j|^p \ge \varepsilon P F^p$, and one uses the following estimate of $\Pr\{(s_r,t_r) \in C_i \Delta C_j\}$ in the preceding string of inequalities:

$$\begin{aligned}
\Pr\{(s_r,t_r) \in C_i \Delta C_j\} &= P_F\left(\frac{|f_i(s_r) - f_j(s_r)|}{2F(s_r)}\right) \\
&\ge \left[P_F\left(\frac{|f_i(s_r) - f_j(s_r)|}{2F(s_r)}\right)^p\right]^{1/p} \\
&= \left(\frac{P|f_i - f_j|^p}{2^p P F^p}\right)^{1/p} \ge \frac{\varepsilon^{1/p}}{2}. \quad \blacksquare
\end{aligned}$$

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It follows from the preceding theorem that if $\mathcal{F}$ is VC subgraph, then, for any probability measure P on $(S,\mathcal{S})$ and $\tau \leq \|F\|_{L^2(P)}$,

$$\log 2N(\mathcal{F},L^2(P),\tau) \leq \log(2m_{v,w}) \vee \log\left(\frac{2^{3+1/(v-1)+1/w}\|F\|^2_{L^2(P)}}{\tau^2}\right)^w \leq 2w\log\left(\frac{A\|F\|_{L^2(P)}}{\tau}\right), \tag{3.235}$$

where $w > v$ and $A = A_{v,w} = \max\left(2^{(3+1/(v-1)+1/w)/2},(2m_{v,w})^{1/2w}\right)$.

As this observation shows, Theorem 3.6.9 is precisely what makes VC subgraph classes of functions useful in empirical process theory and practice. One can combine VC subgraph classes of functions by, for example, addition to obtain classes that are no longer VC subgraph but whose covering numbers still enjoy uniform bounds of the same type, and they are equally useful. For example, if $\mathcal{G}$ and $\mathcal{G}$ are VC subgraph, then so are $\mathcal{F}+\mathcal{G} = \{f+g : f \in \mathcal{F}, g \in \mathcal{G}\}$ and $\mathcal{F}-\mathcal{G}$ (if f_i and g_j are the centres of balls of radius ε covering, respectively, $\mathcal{F}$ and $\mathcal{G}$, then the balls with centres f_i+g_j and radius 2ε cover $\mathcal{F}+\mathcal{G}$). This justifies the following definition:

Definition 3.6.10 A class of measurable functions is of VC type with respect to a measurable envelope F of $\mathcal{F}$ if there exist finite constants A, v such that for all probability measures Q on $(S,\mathcal{S})$

$$N(\mathcal{F},L^2(Q),\varepsilon\|F\|_{L^2(Q)}) \leq (A/\varepsilon)^v.$$

Next, we consider an example of VC subgraph classes of functions that is particularly relevant in density estimation. To motivate it, let us anticipate that a class of functions that naturally arises in the analysis of density estimators based on convolution kernels is

$$\mathcal{K} = \{K((t-\cdot)/h) : t \in \mathbb{R}, h > 0\},$$

where K is a function of bounded variation, and that, in the case of wavelet density estimators, the corresponding class of functions (projection kernels) is

$$\mathcal{F}_\phi = \left\{\sum_{k\in\mathbb{Z}} \phi(2^j y-k)\phi(2^j(\cdot)-k) : y \in \mathbb{R}, j \in \mathbb{N}\cup\{0\}\right\},$$

where ϕ is an α-Hölder continuous function with bounded support for some $\alpha \in (0,1]$ (ϕ is the scaling function of a Daubechies wavelet). Many properties of convolution kernel or of wavelet projection density estimators require good estimates on the size of the empirical process indexed by each of these classes, and these estimates follow as a direct consequence of the fact that these classes of functions are of VC type. We now prove that this is indeed the case. It is convenient to recall the following classical definition: given $1 \leq p < \infty$, a function $f : \mathbb{R} \mapsto \mathbb{R}$ is of *bounded p-variation* if the quantity

$$v_p(f) := \sup\left\{\sum_{i=1}^n |f(x_i)-f(x_{i-1})|^p : -\infty < x_0 < \cdots < x_n < \infty, n \in \mathbb{N}\right\}$$

is finite. In this case, the total p-variation of f is defined as $v_p(f)$, and the p-variation function of f is defined as $v_{p,f}(x) = v_p(fI_{(-\infty,x]})$, $x \in \mathbb{R}$. The functions of bounded 1-variation are precisely the functions of bounded variation. Note also that if f is α-Hölder continuous

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with compact support, then f is of bounded $1/\alpha$-variation. These functions are relevant in density estimation because most convolution kernels are of bounded variation, and the scaling functions of Daubechies wavelets, being α-Hölder for some $\alpha \in (0,1]$, are functions of bounded $(1/\alpha)$-variation (see Chapter 4).

Functions of bounded p-variation admit the following decomposition:

Lemma 3.6.11 *Let f be a function of bounded p-variation. Then $f = g \circ h$, where h is nondecreasing and $0 \le h(x) \le v_p(f)$, and g is $1/p$-Hölder continuous on the interval $[0, v_p(f)]$ and $\|g\|_\infty = \|f\|_\infty$.*

Proof Set $h = v_{p,f}$, and let $R_h \subseteq [0, v_p(f)]$ denote its range. By definition, for any $x < y$, $|f(y) - f(x)|^p \le h(y) - h(x)$, showing that f is constant on the level sets $h^{-1}\{u\}$ of h for any $u \in R_h$. For $u \in R_h$, define $g(u)$ as the value of f on any of the points of $h^{-1}\{u\}$. Then, for $u, v \in R_h$ and x, y such that $h(x) = u$ and $h(y) = v$, $|g(u) - g(v)| = |f(x) - f(y)| \le |h(x) - h(y)|^{1/p} = |u - v|^{1/p}$. Thus, g is $1/p$-Hölder continuous on $R_h \subseteq [0, v_p(f)]$, and $\|g\|_\infty = \|f\|_\infty$. Then, by the Kirszbraun-McShane extension theorem (see Exercise 3.6.13), g admits an extension to $[0, v_p(f)]$ with the same modulus of continuity and uniform bound. By construction, $f = g \circ h$. ■

It is easy to see that the set of dilations and translations of a nondecreasing function is a *VC*-type class of functions, and as a consequence of the preceding decomposition, this property is also shared by functions of bounded p-variation. This is the content of the next proposition.

Proposition 3.6.12 *Let f be a function of bounded p-variation, $p \ge 1$. Then the collection $\mathcal{F}$ of translations and dilations of f*

$$\mathcal{F} = \{x \mapsto f(tx - s) : t > 0, s \in \mathbb{R}\}$$

is of VC type; concretely, for all $0 < \varepsilon \le v_p^{1/p}(f)$ and $w > 6$, there exists $A_{w,p} < \infty$ such that

$$N\left(\mathcal{F}, L^2(Q), \varepsilon v_p^{1/p}(f)\right) \le \left(\frac{A_{w,p}}{\varepsilon}\right)^{(p \vee 2)w}, \quad 0 < \varepsilon \le 1.$$

If, moreover, f is right (or left) continuous, then $\mathcal{F}$ is of VC type, and the $L^2(Q)$ ε-covering numbers of $\mathcal{F}$ admit the uniform bound $(A_{w,p}/\varepsilon)^{(p \vee 2)w}$ for any $w > 3$.

Proof Assume that f is right continuous, and set $M^p = v_p(f)$. By Lemma 3.6.11, $f = g \circ h$, where g is $1/p$-Hölder continuous on $[0, M^p]$ and h is nondecreasing, right continuous (see Exercise 3.6.16) and $0 \le h(x) \le M^p$ for all $-\infty < x < \infty$. Then, by Exercise 3.6.15, letting h^{-1-} denote the left-continuous generalised inverse of h, we have that for every $s \in \mathbb{R}$ and $t > 0$,

$$\begin{aligned}\{(x,u) \in \mathbb{R} \times [0, M^p] : u \le h(tx - s)\} &= \{(x,u) \in \mathbb{R} \times [0, M^p] : h^{-1-}(u) \le tx - s\} \\ &= \{(x,u) \in \mathbb{R} \times [0, M^p] : h^{-1-}(u) - tx + s \le 0\} \subseteq \mathcal{G},\end{aligned}$$

where $\mathcal{G}$ is the *negativity set* of the vector space of real functions on $\mathbb{R} \times [0, M^p]$ spanned by the three functions $(x,u) \mapsto h^{-1-}(u)$, $(x,u) \mapsto x$ and 1. In particular, $\mathcal{G}$ is a *VC* class of index 4 by Proposition 3.6.6. Then the class $\mathcal{C}_0 = \{\{(x,u) : 0 \le u \le h(tx - s)\} : t > 0, s \in \mathbb{R}\}$ is also

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VC of index at most 4, and consequently, the same is true, as is easy to check, of the class $\mathcal{C}_0 = \{\{(x,u) : -\infty < u \le h(tx-s)\} : t > 0, s \in \mathbb{R}\}$. Therefore, the class of functions

$$\mathcal{M} := \{x \mapsto h(tx-s) : t > 0, s \in \mathbb{R}\}$$

is VC subgraph of index at most 4.

Let $1 \le p \le 2$. Then $2/p \ge 1$, and the first inequality in Theorem 3.6.9 gives that there exists $A_{w,p}$ such that for every Borel probability measure Q on $\mathbb{R}$,

$$N(\mathcal{M}, L^{2/p}(Q), \varepsilon M^p) \le \left(\frac{A_{w,p}}{\varepsilon^{2/p}}\right)^w, \quad 0 < \varepsilon \le 1,$$

for all $w > 3$ (in fact, for $w = 3$, see the notes at the end of the chapter). Now, since g is $1/p$-Hölder continuous, we have that if $m_1, m_2 \in \mathcal{M}$ and $\|m_1 - m_2\|_{L^{2/p}(Q)} \le \tau$, then

$$\left[\int (g(m_1) - g(m_2))^2 dQ\right]^{1/2} \le \left[\int |m_1 - m_2|^{2/p} dQ\right]^{1/2} \le \tau^{1/p},$$

showing that any τ-covering of $\mathcal{M}$ for the $L^{2/p}(Q)$ distance induces a $\tau^{1/p}$ covering of $\mathcal{F}$ in $L^2(Q)$. Combining this observation with the preceding estimate on the covering numbers of $\mathcal{M}$, we obtain

$$N\left(\mathcal{F}, L^2(Q), \varepsilon M\right) \le \left(\frac{A_{w,p}}{\varepsilon^2}\right)^w.$$

The case $p \ge 2$ follows in an analogous way, the only difference being that now one uses the second inequality in Theorem 3.6.9, valid for $L^{2/p}(Q)$-covering numbers with $2/p < 1$. This proves the proposition for f right continuous and, by analogy, for f left continuous (just use right-continuous inverses).

Without continuity assumptions, using part (b) of Exercise 3.6.15, one obtains

$$\{(x,u) \in \mathbb{R} \times [0, M^p] : u \le h(tx-s)\} = \{(x,u) \in \mathbb{R} \times [0, M^p] : h^{-1-}(u) \le tx - s, u \in R_1\}$$
$$\times \cup \{(x,u) \in \mathbb{R} \times [0, M^p] : h^{-1-}(u) < tx - s, u \in R_2\}$$

for a convenient partition $\{R_1, R_2\}$ of $[0, M^p]$, and the arguments in the first part of the proof then show that $\mathcal{M}$ is VC subgraph, but now the collection of its subgraphs is the union of two VC classes each of index bounded by 4, that is, by the proof of Proposition 3.6.7, for a VC class $\mathcal{V}$ such that $m^{\mathcal{V}}(n) \le cn^6$ for some $c < \infty$ and all $n \in \mathbb{N}$. Then Theorem 3.6.9 gives, for example, in the case $2/p \ge 1$, $N(\mathcal{M}, L^{2/p}(Q), \varepsilon M^p) \le A_{w,p}/\varepsilon^{2w}$, for any $w > 6$. ■

3.6.3 VC Hull and VC Major Classes of Functions

Other types of classes of functions related to the VC property, but with sensibly larger yet still manageable uniform bounds for their $L^2(Q)$-metric entropies, are the VC hull and the VC major classes. The result developed in this section shows that if a class of functions admits a bound on its $L^2(Q)$ covering numbers of the order ε^w uniformly in Q, then the covering numbers of its convex hull admit a uniform bound of the order $e^{\varepsilon^{-2w/(2+w)}}$.

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Definition 3.6.13 Given a class of functions $\mathcal{F}$, $\mathrm{co}(\mathcal{F})$ is defined as the convex hull of $\mathcal{F}$, that is,

$$\mathrm{co}(\mathcal{F}) = \left\{ \sum_{f\in\mathcal{F}} \lambda_f f : f \in \mathcal{F}, \sum_f \lambda_f = 1, \lambda_f \geq 0, \lambda_f \neq 0 \text{ only for finitely many } f \right\},$$

and $\overline{\mathrm{co}}(\mathcal{F})$ is defined as the pointwise sequential closure of $\mathrm{co}(\mathcal{F})$, that is, $f \in \overline{\mathrm{co}}(\mathcal{F})$ if there exist $f_n \in \mathrm{co}(\mathcal{F})$ such that $f_n(x) \to f(x)$ for all $x \in S$ as $n \to \infty$. If the class $\mathcal{F}$ is *VC* subgraph, then we say that $\overline{\mathrm{co}}(\mathcal{F})$ is a *VC* hull class of functions.

Example 3.6.14 Let $\mathcal{F}$ be the class of monotone nondecreasing functions $f : \mathbb{R} \mapsto [0,1]$. Then $\mathcal{F} \subseteq \overline{\mathrm{co}}(\mathcal{G})$, where $\mathcal{G} = \left\{ I_{(x,\infty)}, I_{[x,\infty)} : x \in \mathbb{R} \right\}$. To see this, just note that if

$$f_n = \frac{1}{n}\sum_{i=1}^{n-1} I_{\{f>i/n\}} = \sum_{j=0}^{n-1} \frac{j}{n} I_{\{j/n<f\leq(j+1)/n\}},$$

then $\sup_{x\in\mathbb{R}} |f_n(x) - f(x)| \leq 1/n$ and that the sets $\{f > i/n\}$ are half-lines, so $I_{\{f>i/n\}} \in \mathcal{G}$.

The *VC* major classes constitute a generalisation of this example: $\mathcal{F}$ is a *VC major class* if the collection of sets $\{f \geq t\}$ for $t \in \mathbb{R}$ and $f \in \mathcal{F}$ is a *VC* class of sets. Proceeding as in the preceding example, we see that if $\mathcal{F}$ is *VC* major and the functions in $\mathcal{F}$ take values in $[0,1]$, then $\mathcal{F}$ is *VC* hull. If $\mathcal{F}$ is just uniformly bounded, then it is a multiple of a *VC* hull class. We will not consider these classes any further.

Now we prove the main result on *VC* hull classes. We begin with a fundamental lemma that estimates the covering numbers of convex hulls of finite classes in terms of their cardinality and their diameter.

Lemma 3.6.15 *Let Q be a probability measure on $(S,\mathcal{S})$, and let $\mathcal{F} = \{f_1,\ldots,f_n\}$ be a collection of n functions in $\mathcal{L}^2(Q)$. Then, for all $\varepsilon > 0$,*

$$N(\mathrm{co}(\mathcal{F}), L^2(Q), \varepsilon(\mathrm{diam}\,\mathcal{F})) \leq (e + en\varepsilon^2)^{2/\varepsilon^2}.$$

Proof Given $f \in \mathrm{co}(\mathcal{F})$, let $\lambda_1,\ldots,\lambda_n$ be nonnegative numbers adding up to one such that $f = \sum_{j=1}^n \lambda_j f_j$, and let λ be the discrete probability measure on $\mathcal{F}$ that assigns mass λ_j to $f_j, j = 1,\ldots,n$. Let $Y_1,\ldots,Y_k$ be independent $\mathcal{F}$-valued random variables with common law λ, that is, such that $\Pr\{Y_i = f_j\} = \lambda_j$, $j = 1,\ldots,n$. Then, letting E denote expectation with respect to Pr, we have $f = EY_1$, and moreover, if we set $\bar{Y} = (1/k)\sum_{i=1}^k Y_i$, we obtain, using Fubini's theorem and independence,

$$\begin{aligned} E\|\bar{Y} - f\|^2_{L^2(Q)} &= \int E\left(\frac{1}{k}\sum_{i=1}^k (Y_i - EY_i)\right)^2 dQ = \frac{1}{k^2}\int \sum_{i=1}^k E(Y_i - EY_i)^2 dQ \\ &= \frac{1}{k} E\|Y_1 - EY_1\|^2_{L^2(Q)} \leq \frac{1}{k}(\mathrm{diam}(\mathcal{F}))^2. \end{aligned}$$

Hence, at least one realization of $\bar{Y}$ must be at $L^2(Q)$-distance not exceeding $(\mathrm{diam}(\mathcal{F}))/\sqrt{k}$ from f. Now, independently of λ and hence of $f \in \mathrm{co}(\mathcal{F})$, every such realisation has the form $\sum_{i=1}^k g_i/k$, $g_i \in \mathcal{F}$. These sums only depend on the number x_j of $g_i = f_j$, for $j = 1,\ldots,n$;

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that is, the number of different such averages does not exceed the number of nonnegative integer solutions of the equation $x_1 + \cdots + x_n = k$, namely, $\binom{k+n-1}{k}$. Therefore,

$$N\left(\mathrm{co}(\mathcal{F}), L^2(Q), \mathrm{diam}(\mathcal{F})/\sqrt{k}\right) \le \binom{k+n-1}{k}.$$

Estimating $\binom{n+k}{k}$ by means of Stirling's formula (3.230) and using that $(1+k/n)^n < e^k$ for all $n \in \mathbb{N}$, it follows that $\binom{k+n-1}{k} \le e^k(1+n/k)^k$. This proves the lemma for $\varepsilon \le 1$ just by taking k to be the smallest integer such that $k^{-1} \le \varepsilon^2$ in the last inequality. For $\varepsilon > 1$, the covering number is 1, and the lemma holds as well. ■

The next lemma gives the first step of an induction procedure that will yield the result. The proof of this lemma is itself by induction, and Lemma 3.6.15 plays a basic role in it.

Lemma 3.6.16 *Let Q be a probability measure on $(S, \mathcal{S})$, and let $\mathcal{F}$ be a collection of measurable functions with envelope $F \in L^2(Q)$ such that*

$$N(\mathcal{F}, L^2(Q), \varepsilon \|F\|_{L^2(Q)}) \le C\varepsilon^{-w}, \quad 0 < \varepsilon \le 1.$$

Set $u = 1/2 + 1/w$ and $L = C^{1/w}\|F\|_{L^2(Q)}$. For each $n \in \mathbb{N}$, let $\mathcal{F}_n$ be a maximal $Ln^{-1/w}$-separated subset of $\mathcal{F}$ for the $L^2(Q)$-norm. Then there exists $C_1 < \infty$ depending only on C and w such that

$$\log N(\mathrm{co}(\mathcal{F}_n), L^2(Q), C_1 L n^{-u}) \le n, \quad n \in \mathbb{N}. \tag{3.236}$$

Proof The proof is by induction on n. Given n_0 fixed, the entropy in the statement is zero for all $n \le n_0$ as soon as C_1 satisfies $C_1 C^{1/w} n_0^{-u} \ge 2$. n_0 and C_1 satisfying this condition will be specified later. Let now, with these choices, $n > n_0$ and $m = n/d$, for $d > 1$ large enough, also to be conveniently chosen later (d will slightly depend on n, just enough to ensure that m is an integer). For each $f \in \mathcal{F}_n$ we choose one and only one function $\pi_m f \in \mathcal{F}_m$ at $L^2(Q)$-distance at most $Lm^{-1/w}$ from f: $\pi_m f$ exists by the definition of $\mathcal{F}_m$. Then the decomposition $f = \pi_m f + (f - \pi_m f)$ induces a decomposition of any $g = \sum_{f \in \mathcal{F}_n} \lambda_f f \in \mathrm{co}(\mathcal{F}_n)$, where $\sum \lambda_f = 1$ and $\lambda_f > 0$, as

$$g = \sum_{f \in \mathcal{F}_m} \mu_f f + \sum_{f \in \mathcal{F}_n} \lambda_f (f - \pi_m f), \tag{3.237}$$

where $\mu_f \ge 0$ and $\sum \mu_f = 1$.

By definition of L and the hypothesis, the cardinality of $\mathcal{F}_n$ does not exceed n. Hence, the set of functions $\mathcal{G}_n = \{f - \pi_m f : f \in \mathcal{F}_n\}$ has cardinality at most n. Moreover, since $\|f - \pi_m f\|_{L^2(Q)} \le Lm^{-1/w}$, the diameter of $\mathcal{G}_n$ is dominated by $2Lm^{-1/w}$. Then, if we apply Lemma 3.6.15 to $\mathcal{G}_n$ with ε such that $2m^{-1/w}\varepsilon = (1/2)C_1 n^{-u}$, it follows that we can cover $\mathrm{co}(\mathcal{G}_n)$ by a collection of balls of radius at most $2^{-1}C_1 L n^{-u}$ whose cardinality does not exceed $\left(e + eC_1^2/16d^{2/w}\right)^{32d^{2/w}C_1^{-2}n}$. Let $\mathcal{K}_1$ be the collection of these centres.

The induction hypothesis on $\mathcal{F}_m$ implies that there exists a covering of $\mathrm{co}(\mathcal{F}_m)$ consisting of (at most) e^m balls of radius at most $C_1 L m^{-u}$. Since $\mathcal{F}_m$ has m elements, its linear span is a subspace H_m of $L^2(Q)$ of dimension at most m; hence, each of the e^m balls of the covering of $\mathrm{co}(\mathcal{F}_m)$ is in fact a ball of radius at most $C_1 L m^{-u}$ in an m-dimensional Hilbert space. By, for example, Exercise 3.6.5, each such ball admits a covering by balls of radius $C_1 L n^{-u}/2$ of

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cardinality at most $\left((3C_1Lm^{-u})/(C_1Ln^{-u}/2)\right)^m = (6d^u)^{n/d}$. Let $\mathcal{K}_2$ be the union of the centres of the balls of these 2^m coverings.

Then, by (3.237), the collection of balls with centres at the functions in the set $\mathcal{K} := \{f+g : f \in \mathcal{K}_1, g \in \mathcal{K}_2\}$ and radius C_1Ln^{-u} cover $\mathrm{co}(\mathcal{F}_n)$, and by the considerations in the preceding two paragraphs, the cardinality of this cover is at most

$$e^{n/d}(6d^u)^{n/d}\left(e+\frac{eC_1^2}{16d^{2/w}}\right)^{32d^{2/w}C_1^{-2}n} = \exp\left[n\left(\frac{1+u\log(6d)}{d}+\frac{32d^{2/w}}{C_1^2}\left(1+\log\frac{C_1^2}{16d^{2/w}}\right)\right)\right].$$

Now we take d_0 large enough so that $(1+u\log(12d_0))/d_0 \le 1/2$, $n_0 \ge 2d_0$, and C_1 large enough so that both $C_1C^{1/w}n_0^{-u} \ge 2$ and $(64d_0^{2/w}/C_1^2)\left(1+\log(C_1^2/16d_0^{2/w})\right) \le 1/2$ (for this last choice, note that $x^{-1}\log(1+x) \to 0$ as $x \to \infty$, decreasing for $x \ge 2$). With these choices, for each $n \ge n_0$, we take $d \in [d_0, 2d_0]$ such that $n/d \in \mathbb{N}$ (which is possible because $n/d_0 - n/2d_0 \ge n_0/d_0 - n_0/2d_0 \ge 1$). For these $d = d_n$, C_1 and $n \ge n_0$, the cardinality of the cover of $\mathrm{co}(\mathcal{F}_n)$ just constructed is at most e^n, which completes the induction argument (note that the choice of C_1 and n_0 ensure the validity of (3.236) for $n \le n_0$). ■

We are now ready to prove the following.

Theorem 3.6.17 *Let Q be a probability measure on $(S,\mathcal{S})$, and let $\mathcal{F}$ be a collection of measurable functions with envelope $F \in L^2(Q)$ such that*

$$N(\mathcal{F}, L^2(Q), \varepsilon\|F\|_{L^2(Q)}) \le C\varepsilon^{-w}, \quad 0 < \varepsilon \le 1. \tag{3.238}$$

Then there exists a constant K depending only on C and w such that

$$\log N(\overline{\mathrm{co}}(\mathcal{F}), L^2(Q), \varepsilon\|F\|_{L^2(Q)}) \le K\varepsilon^{-2w/(w+2)}, \quad 0 < \varepsilon \le 1. \tag{3.239}$$

Proof Suppose that the theorem holds for finite collections of functions satisfying (3.238), and let $\mathcal{F}$ satisfy (3.238). Then, given $0 < \varepsilon < 1$, there exists a $\varepsilon\|F\|_{L^2(Q)}$-dense subset $\mathcal{G}$ of $\mathcal{F}$ that is finite. $\mathcal{G}$ obviously satisfies (3.238) and therefore also (3.239). But by convexity of the $L^2(Q)$-norm, any covering of $\mathrm{co}(\mathcal{G})$ by balls of radius $\varepsilon\|F\|_{L^2(Q)}$ (or less) induces a covering of $\mathrm{co}(\mathcal{F})$ by balls of radius $2\varepsilon\|F\|_{L^2(Q)}$ (and the same centres) so that, for this ε,

$$\log N(\overline{\mathrm{co}}(\mathcal{F}), L^2(Q), \varepsilon\|F\|_{L^2(Q)}) \le N(\mathrm{co}(\mathcal{G}), L^2(Q), \varepsilon\|F\|_{L^2(Q)}) \le K2^{w/(w+2)}\varepsilon^{-2w/(w+2)}.$$

Thus, we may assume that $\mathcal{F}$ is finite.

Set $u = (w+2)/2w = 1/2 + 1/w$ and $L = C^{1/w}\|F\|_{L^2(Q)}$. By assumption for $(C/n)^{1/w} < 1$ and trivially for $(C/n)^{1/w} \ge 1$, $\mathcal{F}$ can be covered by n or fewer balls of radius at most $Ln^{-1/w}$, and we let $\mathcal{F}_n$ denote the collection of the centres of such a covering, $n \in \mathbb{N}$. In particular, for each n, $\mathcal{F}_n$ consists of at most n functions. The theorem will be proved if we show that there exist constants C_k, D_k such that $\sup_{k\in\mathbb{N}} \max(C_k, D_k) < \infty$, and $q > 1$, satisfying

$$\log N(\mathrm{co}(\mathcal{F}_{nk^q}), L^2(Q), C_kLn^{-u}) \le D_kn, \quad n,k \ge 1. \tag{3.240}$$

(Note that given n, there exists $k < \infty$ such that $\mathcal{F}_{nk^q} = \mathcal{F}$.)

Lemma 3.6.16 proves (3.240) for $k = 1$ and all n with $C_1 < \infty$ and $D_1 = 1$. To proceed by induction, we assume that (3.240) holds for $k-1$ and all n and for $q \ge 3w$. Proceeding as in the proof of Lemma 3.6.16, we have

$$\mathrm{co}(\mathcal{F}_{nk^q}) \subseteq \mathrm{co}(\mathcal{F}_{n(k-1)^q}) + \mathrm{co}(\mathcal{G}_{n,k}),$$

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where $\mathcal{G}_{n,k}$ is a collection of at most nk^q functions of $L^2(Q)$-norm at most $L(n(k-1)^q)^{-1/w}$. Thus, Lemma 3.6.15 applied to $\mathcal{G}_{n,k}$ for $\varepsilon = Lk^{-2}n^{-u}/(2L(n(k-1)^q)^{-1/w})$ shows the existence a covering of $\mathcal{G}_{n,k}$ of cardinality at most $(e+ek^{2q/w+q-4}/4)^{2^{3+2q/w}k^{4-2q/w}n}$ by balls of radius not larger than $Lk^{-2}n^{-u}$. The induction hypothesis applied to $\mathcal{F}_{n(k-1)^q}$ yields the existence of a covering of $\mathrm{co}(\mathcal{F}_{n(k-1)^q})$ of cardinality at most $C_{k-1}Ln^{-u}$ by balls of radius no larger than $e^{D_{k-1}n}$. Then the sums $f+g$ of centres f from the balls covering $\mathcal{G}_{n,k}$ and centres g from the balls covering $\mathrm{co}(\mathcal{F}_{n(k-1)^q})$ are the centres of at most

$$\exp\left[n\left(D_{k-1}+\frac{2^{3+2q/w}\log(1+k^{2q/w+q-4}/4}{k^{2q/w-4}}\right)\right]$$

balls of radius at most $C_{k-1}Ln^{-u}+Lk^{-2}n^{-u}$ covering $\mathrm{co}\mathcal{F}_{nk^q}$. Thus, inequality (3.240) is proved for k and for all n with C_k and D_k given by

$$C_k = C_{k-1}+\frac{1}{k^2} \quad \text{and} \quad D_k = D_{k-1}+\frac{2^{3+2q/w}\log(1+k^{2q/w+q-4}/4}{k^{2q/w-4}},$$

which satisfy $\sup_{k\in\mathbb{N}}\max(C_k,D_k)<\infty$ given that $q\geq 3w$. ■

Exercises

3.6.1 Prove that: $\nu(\mathcal{C})=0$ if and only if $\mathcal{C}=\emptyset$. $\nu(\mathcal{C})=1$ if and only if $\mathcal{C}$ consists of only one set. *Hint*: $\mathcal{C}$ shatters the empty set if and only if $\mathcal{C}$ contains at least one set. If $\mathcal{C}$ contains two different sets A and B, then it shatters any set $\{x\}$ for $x\in A\Delta B\neq\emptyset$.

3.6.2 Prove that: If S is an infinite set and $\mathcal{C}$ is the collection of all subsets of S of cardinality k, then all sets of cardinality not exceeding k are shattered by $\mathcal{C}$, but no set of cardinality $k+1$ is shattered. This implies that the bound for $m^{\mathcal{C}}(n)$ in Theorem 3.6.2 is attained.

3.6.3 Prove that: If $\mathcal{C}$ is ordered by inclusion, then $\nu(\mathcal{C})=2$: $\mathcal{C}$ cannot shatter any set consisting of two distinct points. The same is true if $\mathcal{C}$ consists of disjoint sets.

3.6.4 Let S and T be sets, and let $F: S\mapsto T$ be a function. Let $\mathcal{C}$ be a collection of subsets of T, and let $F^{-1}(\mathcal{C})=\{F^{-1}(C): C\in\mathcal{C}\}$. Prove that $\nu(F^{-1}(\mathcal{C}))\leq\nu(\mathcal{C})$.

3.6.5 Let $B(a,r)=\{x\in\mathbb{R}^d: |x-a|\leq r\}$ be a ball of radius R and centre a in $\mathbb{R}^d$. Prove the following bound for the packing number of $B(a,r)$ with respect to Euclidean distance $d(x,y)=|x-y|$, that is,

$$D(B(a,r),d,\varepsilon)\leq\left(\frac{3r}{\varepsilon}\right)^d, \quad 0<\varepsilon\leq r.$$

Hint: See Proposition 4.3.34.

3.6.6 Let $\mathcal{F}$ be a class of (measurable) functions ordered by the relation $f\leq g$ iff $f(x)\leq g(x)$, for all $x\in S$, and assume that $0\leq f\leq 1$ for all $f\in\mathcal{F}$. Show that $N(\mathcal{F},L^2(Q),\varepsilon)\leq 2\varepsilon^{-2}$ for all $0<\varepsilon\leq 1$ and probability measures Q on $(S,\mathcal{S})$. *Hint*: Each set in the partition of $\mathcal{F}$ $A_k=\{(k-1)\varepsilon^2\leq f<k\varepsilon^2: f\in\mathcal{F}\}$, $1\leq k\leq\varepsilon^{-2}+1$, is contained in a $L^2(Q)$-ball of radius at most ε (for $g\leq f$ both in A_k, $Q(f-g)^2\leq Q(f-g)\leq\varepsilon^2$). *Note*: Theorem 3.6.9 falls short of implying this simple result (it gives a bound for the covering numbers of the order of ε^{-w} for any $w>2$; see the notes at the end of the chapter).

3.6.7 Use Exercise 3.6.6, Example 3.6.14 and Theorem 3.6.17 to show that if $\mathcal{F}$ is the class of monotonically nondecreasing functions $f:\mathbb{R}\mapsto[0,1]$, then there exist $K<\infty$ such that

$$\log N(\mathcal{F},L^2(Q),\varepsilon)\leq K/\varepsilon, \quad 0<\varepsilon<1.$$

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Show that the same is true for the class of monotonically nondecreasing functions and for the class of functions of bounded variation on $\mathbb{R}$ taking values on taking values on $[0,1]$ and of variation bounded by one. [See also Corollary 3.7.50]. *Hint*: Recall that functions of bounded variation are differences of monotone functions.

3.6.8 If $\mathcal{F}$ is countable and $PF < \infty$, where $F = \sup_{f\in\mathcal{F}} |f|$, then $\|P_n - P\|_{\mathcal{F}}$ converges almost surely and in L^1. *Hint*: Assume that the variables X_i that make the empirical process are the coordinate functions on an infinite product probability space $(S^{\mathbb{N}}, \mathcal{S}^{\mathbb{N}}, P^{\mathbb{N}})$. Let $\mathcal{S}_n$ be the smallest σ-algebra that contains the sets of $P^{\mathbb{N}}$-measure zero and the sets in

$$\{A \in \mathcal{S}^{\mathbb{N}} : I_A(x) = I_A(\sigma_n x) \text{ for any permutation } \sigma_n \text{ of the first } n \text{ coordinates}\}.$$

Then, if $A \in \mathcal{S}_n$ and $i,j \le n$, $\int_A f(X_i)d\Pr = \int_A f(X_j)d\Pr = \int_A P_n(f)d\Pr$, and therefore, since, moreover, $\|\cdot\|_{\mathcal{F}}$ is convex,

$$E(\|P_{n-1} - P\|_{\mathcal{F}} | \mathcal{S}_n) \ge \|E((P_{n-1} - P)(f)|\mathcal{S}_n)\|_{\mathcal{F}} = \|P_n - P\|_{\mathcal{F}}.$$

This fact requires no measurability other than that the X_i being the coordinates of a product probability space (see Proposition 3.7.8). Thus, $\|P_n - P\|_{\mathcal{F}}$ is a reverse submartingale if $EF < \infty$. Now apply reverse submartingale convergence.

3.6.9 (Vapnik-Červonenkis Glivenko-Cantelli theorem.) Use the preceding exercise, the bound for the packing numbers of *VC* subgraph classes in this section and Theorem 3.5.6 to show that if $\mathcal{F}$ is a countable *VC* subgraph class of functions such that if $PF < \infty$, then $\|P_n - P\|_{\mathcal{F}} \to 0$ a.s. Deduce that the same is true if $\mathcal{F}$ is *VC* hull or *VC* major (assuming that $PF < \infty$).

3.6.10 Produce versions of Theorem 3.6.17 in the following cases: a) $\overline{\text{co}}\mathcal{F}$, where $\mathcal{F} = \{\sum_{i=1}^k g_i : g_i \in \mathcal{F}_i\}$, where $\mathcal{F}_i$ are *VC* subgraph classes and $k < \infty$; (b) $\mathcal{F}$ just as in (a), but the convex hull of $\mathcal{F}$ is replaced by its *symmetric convex hull* $\overline{\text{sco}}\mathcal{F}$ – same definition except that $\sum_f |\lambda_f| \le 1$ – and (c) for $M\overline{\text{co}}\mathcal{F}$ or $M\overline{\text{sco}}\mathcal{F}$ for any M finite.

3.6.11 Prove that: Any finite dimensional space of functions is *VC* subgraph.

3.6.12 Show that if $\mathcal{F}$ is *VC* subgraph, then so are $\mathcal{F}+g$, $\mathcal{F}\cdot g$, $\mathcal{F}\circ g$, for any function g, and $g\circ\mathcal{F}$ if g is monotone.

3.6.13 Let $K : \mathbb{R}^d \mapsto \mathbb{R}$ be a finite linear combination of measurable functions k whose subgraphs $\{(s,u) : k(s) \ge u\}$ can be represented as a finite number of Boolean operations of sets of the form $\{(s,u) : p(s,u) \ge \phi(u)\}$, where p is a polynomial and ϕ is an arbitrary measurable function. Prove that the collection of functions

$$\left\{K\left(\frac{t-\cdot}{h}\right) : t \in \mathbb{R}^d, h > 0\right\}$$

is *VC* type. Examples: $K(x) = L(\|x\|)$, where L is of bounded variation, and $K = I_{[-1,1]^d}$.

3.6.14 (Kirszbraun-McShane extension theorem.) Let (T,d) be a metric space and $S \subset T$. Let $f : S \mapsto \mathbb{R}$ be bounded by M in absolute value and admit a modulus of continuity φ, that is, $|f(s)| \le M$ for all $s \in S$ and $|f(t) - f(s)| \le \varphi(d(s,t))$ for all $s,t \in S$, where $\varphi : [0,\infty) \mapsto (0,\infty)$ satisfies $\varphi(0) = 0$ and $0 \le \varphi(x) \le \varphi(x+y) \le \varphi(x) + \varphi(y)$ for $x,y \ge 0$. Prove that there exists an extension g of f defined on all of T that admits on T the bound M (for its absolute value) and the modulus of continuity φ. *Hint*: Prove that if $h(t) := \inf_{s\in S}[f(s) + \varphi(d(s,t)]$, $t \in T$, then the function $g(t) = \max[\min(h(t), M), -M]$ satisfies the prescribed properties. (See, e.g., Dudley (2002).)

3.6.15 (Quantile functions.) Let f be a monotone nondecreasing function on $\mathbb{R}$, with $-\infty < a = f(-\infty+) < f(+\infty-) = b$, and let f^{-1-} denote its left-continuous generalised inverse $f^{-1-}(t) = \inf\{x : f(x) \ge t\}$. (a) Prove that if f is right continuous, then, for $x \in \mathbb{R}$ and $t \in (a,b)$, $f(x) \ge t$ if and only if $x \ge f^{-1-}(t)$. (b) Prove that (a) does not hold without the continuity assumption. (c) Show that for any given $t \in (a,b)$, either $\{x : f(x) \ge t\} = \{x : x \ge f^{-1-}(t)\}$ or

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